BANK OF THAILAND

Monetary Policy Report

June 2015

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
June 2015

Mr. Prasarn Trairatvorakul	Chairman
Mrs. Pongpen Ruengvirayudh	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Jamlong Atikul	Member
Mr. Porametee Vimolsiri	Member
Mr. Veerathai Santiprabhob	Member
Mr. Sethaput Suthiwart-Narueput	Member

Monetary Policy in Thailand

The Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the Governor and two deputy Governors, as well as four distinguished external members representing various sectors of the country, with the aim of ensuring that monetary policy decisions are effective and transparent.

The Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

The Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the monetary policy target for 2015, in place of the quarterly average core inflation target of 0.5 – 3.0 percent. The new inflation target was jointly proposed by the MPC and the Minister of Finance. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the public, together with measures taken and estimated time to bring inflation back to the target.

The Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14.00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the Monetary Policy Report every quarter.

Monetary Policy Report

June 2015

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Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy is likely to expand at a slower pace than the previous projection mainly on the back of weaker-than-anticipated exports. Sluggish global economic recovery, changing global trade structure, and more pronounced structural constraints in the Thai exports sector together contributed to the subdued exports. The continued decline in exports adversely affected Thailand's economic recovery through lower income, confidence and private spending. Meanwhile, robust tourism sector and higher public spending, particularly investment expenditure, provided support to the economy, but could not fully offset the impacts of aforementioned negative factors.

Inflationary pressures subsided mainly from supply-side factors, causing headline inflation to remain in a negative territory for longer than previously assessed. However, headline inflation is expected to pick up in the second half of 2015 and move closer to the inflation target in 2016, as the base effects of high oil price begins to wane, coupled with expected rises in oil and raw food prices. In the light of higher economic downside risks and muted inflationary pressure, the MPC lowered the policy rate at the April meeting to support economic recovery.

1.1 Growth and inflation prospects

The economy is projected to recover at a slower pace than assessed in the previous Monetary Policy Report. Thai exports fell short of expectation due to a combination of factors. First, global economic recovery was held back by subdued growth in the U.S., China and Asia. The changing global trade structure also prompted Thailand's major trading partners to rely less on imports (*Article in Box 1: The Changing Global Trade Structure and Implications on Exports of the ASEAN-5 Countries*). Moreover, the Thai exports sector continued to suffer from structural problems.

Falling exports had adverse impacts on employment, incomes and household confidence, thereby offsetting the positive effects of low oil prices and more accommodative monetary conditions. In addition, financial institutions remained cautious in lending to households, contributing to slower-than-expected recovery of household consumption.

Private investment is projected to rebound at a slower pace than previously assessed due to weakening external and internal demands. Feeble demands dampened private sector confidence, prompting firms to delay their new investment until clearer signs of demand recovery emerged. Public spending, especially investment expenditure picked up more strongly than projected, helping to spur investment by private firms benefitting from government investment projects. Meanwhile, tourism is projected to post a solid growth throughout the forecast period, especially in 2015.

However, accelerating fiscal spending and robust tourism growth could not offset the negative impacts from contracting exports and private spending. The MPC thus projected the Thai economy to grow by 3.0 percent and 4.1 percent in 2015 and 2016 respectively **(Table 1.1)**.

Table 1.1 Forecast summary

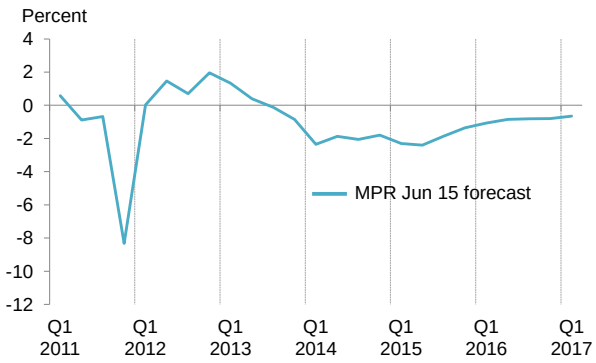
Percent	2014*	2015	2016
GDP growth**	0.9	3.0	4.1
		(3.8)	(3.9)
Headline inflation	1.9	-0.5	1.6
		(0.2)	(2.2)
Core inflation	1.6	1.0	1.0
		(1.2)	(1.2)

Note: * Outturn
** Forecast based on database of GDP-CVM (chain volume measure)
( ) MPR Mar 2015
Source: Office of the National Economic and Social Development Board, Ministry of Commerce and forecast by Bank of Thailand

With inflationary pressures having dropped below the assessment in the last Monetary Policy Report, the Committee revised down headline and core inflation forecasts throughout the forecast period. Headline inflation forecast is adjusted downwards primarily because of supply-side factors, particularly with oil prices remaining at low levels. The reduction of the contribution rate to the Oil Fund caused retail oil prices to increase by less than the rise in global oil prices. Moreover, raw food prices posted a larger drop than projected due to an oversupply of

agricultural products, especially meat and eggs. At the same time, demand-side inflationary pressures weakened in line with slower-than-expected recovery. Negative headline inflation is thus likely to persist longer in 2015. Nevertheless, headline inflation is expected to pick up gradually in the second half of the year and move closer to the inflation target in 2016, on the expectation that the base effects of high oil price will begin to wane while energy and raw food prices will increase. Core inflation remains positive but is projected to decline, in line with weak demand-side pressures resulting from the persistent negative output gap. The MPC considered deflation risks to be low, as private consumption continued to expand, prices of most goods and services remained stable or rose, and inflation expectations were close to the target **(Table 1.1 and Chart 1.1)**.

Chart 1.1 Output Gap



The Committee factored the following key developments into the growth and inflation forecasts.

(1) The global economy is projected to grow at slower rate than formerly forecast, due to a slowdown in the U.S., Chinese and Asian economies, along with shifting global trade structure (Table 1.2).

Throughout the forecast period, the quantity of Thai exports was hit by the slower-than-expected global recovery, along with the recent changes in global trade structure that caused Thailand's main trading partners to reduce their reliance on imports and switch to locally made products. This reduced Thailand's benefits from the global economic recovery. The prices of Thai exports are projected to remain steady at low levels due to soft demand. Moreover, Thai exports suffered from structural constraints. The lack of new investment and product development to meet the demands of the global market lowered

Table 1.2 Growth assumptions for Thailand's trading partners

Annual percentage change (%YoY)	Weight (%)	2014	2015		2016	
			Mar 2015	Jun 2015	Mar 2015	Jun 2015
The U.S.	14.9	2.4	3.2	2.1	2.9	2.9
The euro area	10.0	0.9	1.1	1.3	1.5	1.6
Japan	13.6	0.0	1.0	0.8	1.4	1.4
China	15.7	7.4	7.0	6.9	6.9	6.8
Asia (excluding Japan and China)*	37.4	4.1	4.2	4.0	4.4	4.4
Total**	100	3.5	3.7	3.3	3.9	3.8

Note: * Weighted by each trading partner's share in Thailand's total exports in 2014 (7 countries: Singapore (6.5%), Hong Kong (7.9%), Malaysia (8.0%), Taiwan (2.5%), Indonesia (5.9%), South Korea (2.8%) and the Philippines (3.7%)
** Weighted by each trading partner's share in Thailand's total exports in 2014 (13 countries)

Thailand's competitiveness. Rectifying these structural problems will take time.

Moreover, in recent periods certain groups of Thai exports were affected by exchange rate movements, particularly for goods where Thailand competed with other countries whose currencies had become significantly weaker against the baht. Nonetheless, the weaker baht in 2015 Q2 following the policy rate cut, coupled with the ensuing announcement of BOT's measures on capital flows relaxation, should boost exporters' profits in baht terms and provide additional support for the price-sensitive exports sectors somewhat. However, the positive effects may not be immediately felt because it takes time for foreign buyers to revise their orders. In addition, the prices of Thai exports may have to be lowered in some product categories where Thai exporters have weak negotiation powers over prices, thereby limiting the positive impacts of the weaker baht.

Exports of services posted a solid expansion from the growing number of Chinese tourists, supporting the economy. The lifting of martial law in April 2015 and more apparent signs of recovery in the euro area economies pointed to a brighter outlook for the tourism sector in periods ahead.

The contraction of exports curbed employment in the manufacturing exports sector. Falling household incomes and eroding confidence prompted consumers to be more cautious with spending. Slow economic recovery further weakened business confidence which, combined with sizable excess production capacity, led the private sector to delay investment plans. Meanwhile, sluggish global recovery continued to weigh down commodity prices, especially agricultural prices, and continued

to impact farm incomes and private consumption in the next periods.

In the MPC's view, global economic growth could turn out to be below the base case scenario for the following reasons: (1) The economic slowdown in China could be worse than assessed due to the lower-than-expected impact from government stimulus; (2) China's economic slowdown could weigh down ASEAN economies more than expected; and (3) The recovery of the euro area economies could be slower than expected, should Greece's default cause excessive volatility in financial markets and erode investor confidence.

(2) The government provides additional boost to the economy through public investment.

The Committee revised upwards the assumptions on public spending as the central government managed to accelerate the disbursement of public investment expenditure in early 2015. This was partly because the public sector had entered into contracts for investment projects more quickly and in greater amounts than expected, which helped sustain public investment spending. Moreover, the government announced further extra-budgetary spending under the second phase of the fiscal stimulus. Major investment plans, including water management system and urgent road transport infrastructure development, are likely to provide ongoing boost to the economy. However, there were delays and revisions of plan in some state enterprise investment projects, such as the second phase of the Suvarnabhumi Airport development plan.

Going forward, higher and sustained public spending will underpin private sector confidence and shore up private investment by firms involved in the government projects. The impacts are likely to be more pronounced in late 2015 and 2016 when the

construction of public infrastructure projects is scheduled to begin and larger budget disbursements are expected. Moreover, the water management and road transport development plans consist mostly of small and medium spending projects which can be implemented relatively swiftly. These projects can thus provide an immediate boost to employment and incomes in local areas, thereby helping to revitalize household consumption to some extent.

The Committee assessed the risk of public spending overshooting the baseline projection to be higher than that of public spending undershooting it. The central government might be able to expedite extra-budgetary spending through the water management and road transport development projects more quickly than currently assessed. This will alleviate some of the labor constraints in the construction sector especially in the later part of the forecast period, as well as facilitate the implementation of projects under the normal budgetary process. Nonetheless, the downside risk that budget disbursement capacity might not keep pace with the budget increase requires ongoing monitoring.

(3) The increase in global oil prices in 2015 Q2.

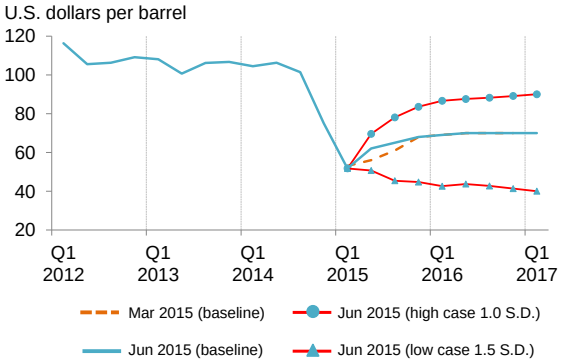
The Committee slightly revised upwards the baseline assumption for crude oil price in 2015 from 60 to 61.7 U.S. dollars per barrel. The revision followed the larger-than-expected increase in oil prices in 2015 Q2, fueled by stronger global demand amidst global economic recovery as well as moderating rate of increase in U.S. oil production from closures of some oil fields. Meanwhile, crude stockpiles began to decline, contributing to the higher oil prices. Looking ahead, however, the rise in crude prices are likely to be limited on the following

grounds: (1) Demand remains soft amidst slow global recovery. (2) Supply should increase as shale oil production in the U.S. resumes, with higher oil prices making production viable again. The MPC therefore maintains the crude price baseline assumption for 2016 at 70 U.S. dollars per barrel **(Chart 1.2)**.

Higher oil prices will put upward pressure on the prices of other commodities that exhibit a high degree of comovements with oil prices, giving a boost to our exports. Moreover, higher crude price will translate into higher domestic retail oil prices, which will in turn push headline inflation up to positive territory in Q4 2015 and subsequently to a level close to the inflation target in 2016.

According to the MPC's evaluation, the crude price could undershoot the baseline assumption, on the back of higher-than-expected oil supply driven by the following factors: (1) Iran could resume exporting oil if the negotiation on the lifting of sanctions is successful. (2) OPEC countries could raise oil production by more than previously assessed in order to maintain market share. (3) Shale oil producers could further expand production due to greater production efficiency. Meanwhile, demand for oil could fall below the expected level, in line with the pace of global recovery which may fall short of the baseline assumption.

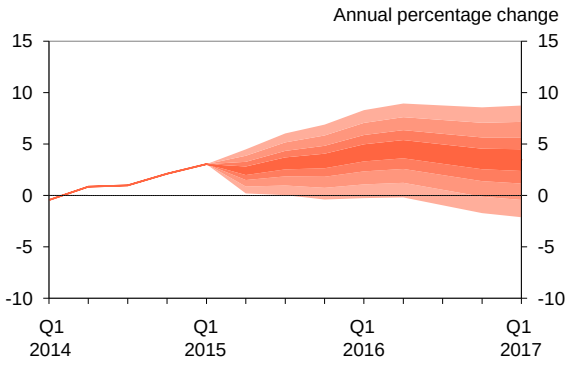
Chart 1.2 Assumptions on Dubai oil price



Downside Risks to Growth and Inflation Forecasts

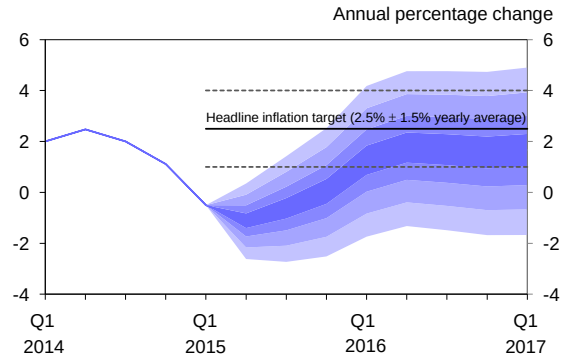
According to the MPC's assessment, **the probability that growth will be below our baseline projection is higher than the probability that it will be above the baseline projection. This assessment is depicted in the growth fan chart, which is skewed downwards throughout the forecast period (Chart 1.3).**

Chart 1.3 GDP growth forecast



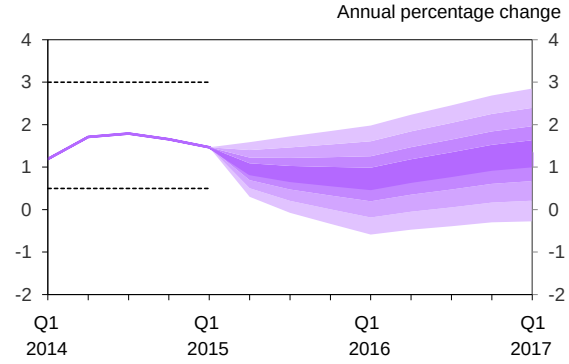
Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.4 Headline inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.5 Core inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

Downside risks to economic growth stem from the following: (1) The pace of global economic recovery could be slower than anticipated, especially in China and Asia. (2) The negative impacts of the shifting global trade structure on Thai exports could be greater than expected. (3) Public spending could fall short of anticipation due to limitations in disbursements, especially for investment projects.

Upside Risk factors to growth could arise from the following sources: (1) Public spending could exceed the previous forecast due to quicker implementation of second-round fiscal stimulus measures, which could in turn spur more private investment. (2) Crude oil prices could be below the baseline projection, providing support to the pick-up in household spending.

The Committee judges that **headline and core inflation are more likely to fall below the central projection than to surpass it. The greater downside risk is reflected in the inflation fan charts that are skewed downwards throughout the forecast period (Charts 1.4 and 1.5).** The assessment is based on the risk factors concerning crude oil prices and the possibility that economic growth would be lower than the baseline projection.

1.2 Monetary policy decision

Monetary policy stance has become more accommodative.

Monetary policy has played a greater role in supporting economic recovery during 2015 Q2, amidst greater downside risks and subdued inflationary pressures. At the meeting on April 29, 2015, the MPC voted to reduce the policy interest rate by 0.25 percent, following the previous rate cut in March.

After the MPC's decision to lower the policy rate by 0.25 percent at the meeting on March 11, 2015, the overall monetary conditions became more accommodative. Commercial banks' deposit and lending rates gradually declined, accompanied by lower government bond yields across all maturities. These adjustments partly eased the financial burden and financing costs faced by the business and household sectors. The baht depreciated for brief periods following the reduction in the policy rate, but subsequently rose due to external factors.

At the meeting on April 29, 2015, the MPC agreed that Thailand's economic growth in 2015 would fall short of the earlier assessment, and that various monetary policy tools must be used to further ease financial conditions. The Committee thus voted 5 to 2 to reduce the policy interest rate by 0.25 percent, from 1.75 percent to 1.50 percent.

The MPC had a thorough deliberation on monetary policy space, in the light of the uncertain prospects of internal and external factors. The Committee considered the effective lower bound of the policy rate to be above zero percent. Should the policy rate move to very low levels, the interest

rate transmission mechanism could become more limited. This is because, even if the policy rate were to fall below the effective lower bound, commercial banks might not be able to cut lending rates any further given the constraints of their asset composition and funding structure.

Furthermore, despite the historically low rate of 1.25 percent in the wake of the global financial crisis in 2009, the Committee judged that the effective lower bound of the policy rate depends on the prevailing economic and financial environment at particular moments. These conditions, such as loan quality outlook, saving alternatives and level of competition in the financial sector, could deviate from the past.

Aside from the above policy deliberation, the MPC discussed the possibility of stronger growth momentum in 2015 Q2. The Committee paid considerable attention to the fiscal boost and the risks posed to Thai exports stemming from the slowdown in Thailand's major trading partners and structural changes in global trade. In the light of these downside risks, the majority of the MPC members deemed a policy rate reduction necessary as a pre-emptive measure to bolster to the economy. They judged that the rate cut would not only reduce the financial burden of business and household sectors, but also lead to exchange rate adjustments that are more conducive to economic recovery. Moreover, given that headline inflation could remain below the lower bound of the target for longer than expected, the policy rate cut would help anchor inflation expectations and ease deflation risks which, although still low, had increased slightly since the previous MPC meeting.

Nonetheless, the minority of the MPC members believed that policy space should be preserved for countering future risks, such as the slowdown of the Chinese economy and domestic political uncertainties. Furthermore, the effects of the policy rate reduction on March 11 had not been fully transmitted to the real economy, highlighting the need to evaluate the impacts on the economy and financial stability before any additional easing monetary policy. Meanwhile, higher fiscal spending and more accommodative monetary conditions should help foster the economic recovery to a certain degree.

The decision to reduce the policy interest rate this time further eased monetary conditions, particularly through the exchange rate adjustment. The baht depreciation was partly a result of the ensuing announcement of BOT's measures on capital flows relaxation on April 30, 2015. The MPC noted and agreed with the BOT's plans to relax foreign exchange regulations under the Capital Account Liberalization Master Plan. The Committee also realized the importance of the BOT's effective communication with the public to promote better understanding of exchange rate movement and its economic impacts. This would help the business sector adapt and manage exchange rate risks more effectively under different circumstances.

Subsequently at the meeting on June 10, 2015, the Committee unanimously decided to maintain the policy rate at 1.50 percent. The MPC believed that the pace of Thailand's economic recovery was still close to the assessment made in the previous meeting. Growth momentum in 2015 Q1 and April softened from feeble private spending and continued contraction in exports, in part a result of a slowdown in the Chinese

and Asian economies as well as a shift in the global trade structure. Nevertheless, improved disbursements of public investment expenditure and robust tourism growth continued to provide support to the economy.

Headline inflation continued to stay in a negative territory due mainly to energy costs and raw food prices. Core inflation remained positive but edged slightly downward due to subdued demand-side pressure. The MPC considered deflation risks to remain low on the grounds that private consumption continued to expand, prices of most goods and services were stable or rose, and inflation expectations remained close to the inflation target.

The Committee judged that the conduct of monetary policy had thus far eased monetary conditions, while the direction of exchange rate movement had become more conducive to the economic recovery. Nonetheless, the Thai economy still faced downside risks, especially those stemming from sluggish global recovery, especially China and other Asian economies. Going forward, the Committee will closely monitor Thailand's economic and financial developments and stand ready to utilize the available policy space appropriately to ensure sufficiently accommodative monetary conditions to foster a recovery of the Thai economy. The MPC will also assess financial imbalances that could build up in a prolonged low interest rate environment, thereby ensuring the country's long-term financial stability.

1.3 Appendix

Table 1.3 Forecasts for GDP and Components*			
Percent (per annum)	2014**	2015	2016
GDP growth	0.9	3.0	4.1
Domestic demand	-0.1	3.1	4.0
Private consumption	0.6	2.0	3.1
Private investment	-2.0	2.7	6.3
Government consumption	1.7	3.3	3.5
Public investment	-4.9	16.3	6.0
Exports of goods and services	0.0	2.2	3.6
Imports of goods and services	-5.4	2.7	4.3
Current Account (billion US dollars)	14.2	19.5	8.3
Value of merchandise exports	-0.3	-1.5	2.5
Value of merchandise imports	-8.5	-2.4	7.6

Note : *Forecast based on database of GDP-CVM (chain volume measure)
** Outturn

Table 1.4 Forecast assumptions			
Annual percentage change	2014	2015	2016
Dubai oil price (U.S. dollars per barrel)	96.7	61.7	70
Non-fuel commodity prices (%YoY)	-3.9	-13.1	1.4
Fresh food prices (%YoY)	4.8	-4.1	3.0
Minimum wage in the Bangkok Metropolitan Region (baht per day)	300	300	300
Government consumption (current price) (%YoY)[1/]	3.4	6.5	6.1
Public investment (current price) (%YoY)[1/]	-4.7	12.8	9.2
Fed Funds rate (% at year-end)	0.13	0.88	2.38
Trading partners' economic growth (%YoY)[2/]	3.5	3.3	3.8
Regional currencies vis-à-vis the U.S. dollar (Index)[3/]	133.5	143.8	144.1

Note : [1/] Including spending on water management plans and infrastructure investment projects
[2/] Weighted by each trading partner's share in Thailand's total exports
[3/] Appreciation against the US dollar indicated by the minus sign

Table 1.5 GDP growth forecasts by research houses		
	2015	2016
Standard Chartered	4.1	4.8
Barclays	3.7	4.5
FPO[1/]	3.7	-
HSBC	3.6	3.1
Maybank Kim Eng	3.5	4.5
JP Morgan	3.5	3.8
NESDB[2/]	3.0-4.0	-
Kiatnakin Bank	3.3	3.7
Phatra	3.3	3.7
DBS Bank	3.2	4.5
Tisco Securities	3.2	4.0
Credit Suisse	3.1	3.8
BOT	3.0	4.1
Capital Economics	3.0	3.5
Kasikorn Research	2.8	3.5
Nomura	2.7	3.5

Note: Compiled and published by Reuters on June 8, 2015, except:
[1/] Published on April 29, 2015
[2/] Published on May 18, 2015, with the release of GDP data for 2015 Q1
Presented in descending order of 2015's forecast

Table 1.6 Headline inflation forecasts by research houses		
	2015	2016
Kasikorn Research	0.5	2.7
Standard Chartered	0.5	2.5
Nomura	0.3	1.2
Credit Suisse	0.3	-
Barclays	0.2	2.5
Kiatnakin Bank	0.2	2.3
Phatra	0.2	2.3
FPO[1/]	0.2	-
NESDB[2/]	(-0.3)-0.7	-
HSBC	0.0	2.6
DBS Bank	-0.2	2.0
BOT	-0.5	1.6
JP Morgan	-0.7	2.0

Note: Compiled and published by Reuters on June 8, 2015, except:
[1/] Published on April 29, 2015
[2/] Published on May 18, 2015, with the release of GDP data for 2015 Q1
Presented in descending order of 2015's forecast

Table 1.7 Probability distribution of GDP growth forecast								
Percent	2015			2016				2017
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 12	0	0	0	0	0	0	0	0
10-12	0	0	0	1	2	2	1	2
8-10	0	0	1	5	7	6	5	6
6-8	0	5	9	16	19	16	14	13
4-6	10	25	27	28	28	25	23	21
2-4	51	42	34	27	25	25	25	24
0-2	35	23	21	15	14	16	18	18
(-2)-0	4	5	6	5	5	7	9	10
< (-2)	0	0	1	1	1	2	4	6

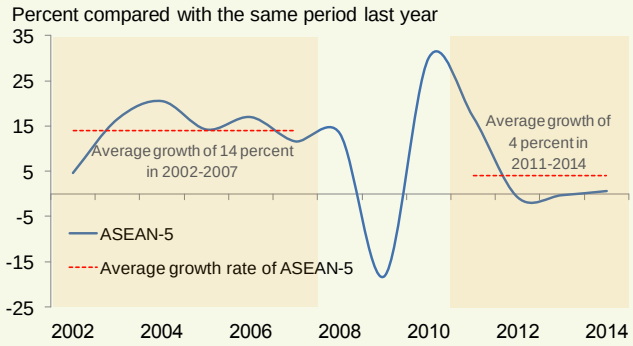
Table 1.8 Probability distribution of headline inflation forecast								
Percent	2015			2016				2017
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
6-7	0	0	0	0	1	1	1	1
5-6	0	0	0	1	3	3	3	3
4-5	0	0	0	4	7	7	7	7
3-4	0	0	2	10	14	13	12	13
2-3	0	2	7	17	20	19	18	18
1-2	1	8	16	22	21	21	20	20
0-1	10	21	24	20	17	17	18	17
(-1)-(0)	34	31	24	14	10	11	12	11
(-2)-(-1)	39	25	16	7	5	5	6	6
(-3)-(-2)	15	11	7	3	2	2	3	3
(-4)-(-3)	2	3	2	1	0	1	1	1

Table 1.9 Probability distribution of core inflation forecast								
Percent	2015			2016				2017
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
3.5-4.0	0	0	0	0	0	0	0	1
3.0-3.5	0	0	0	0	0	1	2	3
2.5-3.0	0	0	0	1	2	3	5	7
2.0-2.5	0	1	3	4	6	9	11	13
1.5-2.0	7	9	10	11	14	17	18	19
1.0-1.5	37	27	23	20	22	22	22	21
0.5-1.0	42	35	29	25	23	21	19	17
0.0-0.5	12	21	22	21	17	15	13	11
(-1)-0.0	1	6	10	12	9	8	6	6
(-2)-(-1)	0	1	3	5	4	3	2	2
< -2	0	0	0	2	1	1	1	1

The changing global trade structure and implications for exports of ASEAN-5 countries[1/]

While the global economy has been recovering since 2013, the benefits of this recovery on exports of ASEAN-5 countries, both in terms of export value and quantity, was relatively low. The total export value of the ASEAN-5 economies grew on average by only 4 percent per year over the 2011-2014 period, compared to the 14-percent annual growth rate during 2002-2007 **(Chart 1)**. The marked decline in export growth could be attributed to the following changes in the global trade structure:

Chart 1 Growth in ASEAN-5's exports value



Source: Trademap

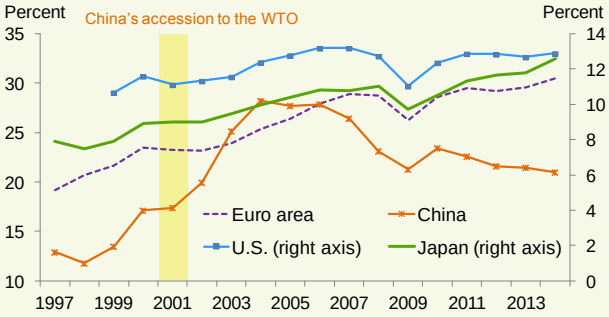
(1) A slowdown in the global trade growth, especially in the aftermath of the financial crisis in 2008-2009. The volume of world trade grew by only 4 percent per year during 2011-2014, moderating from the 7-percent growth per year over the 2002-2007 period. Both cyclical and structural factors have played some roles in the slowdown, including:

(1.1) The global economy remains weak, due to the slowdown of the Chinese economy from ongoing structural reforms, and the sluggish recovery of the euro area and Japanese economies. Moreover, commodities are among the key export products for the ASEAN-5 economies, but global commodity prices remain low.

(1.2) The shift in the import patterns in several countries, particularly the U.S. and China, as reflected in the decreasing imports-to-GDP ratio **(Chart 2)**. The post-crisis imports-to-GDP ratio for the U.S. has fallen from 13.2 percent during 2006-2007 to 12.8 percent during 2011-2014. This can be attributed to **(1) a decrease in oil imports,** thanks to improved drilling technology which leads to increased domestic production of shale gas and shale oil, and **(2) a decrease in imports of consumer products**, which may be attributed, in part, to the rapidly narrowing labor cost gap between the U.S. and developing countries. This argument is striking for China, whose labor cost has been on the rise. Over the 2002-2012 period, the labor cost gap between the U.S. and China has shrunk by about 75 percent. American companies therefore have more incentives to produce onshore **(Chart 3)**.

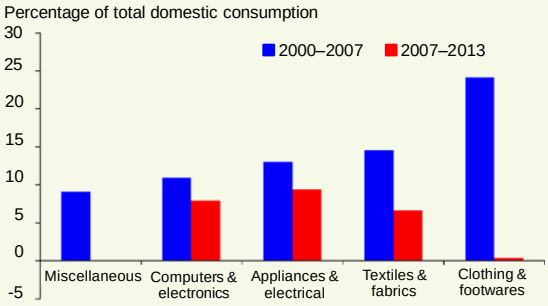
[1/] ASEAN-5 comprises Malaysia, Singapore, Indonesia, the Philippines and Thailand

Chart 2 Imports volume to the size of the real economy ratio



Source: CEIC, Oxford Model

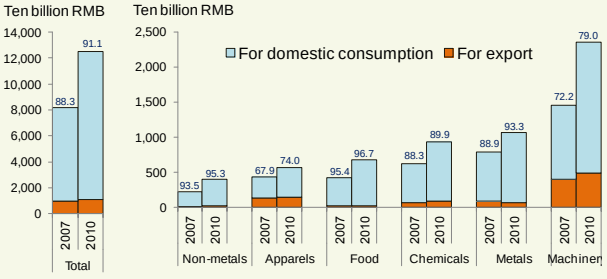
Chart 3 Ratio of relocation of U.S. production base to other countries in various sectors



Source: U.S. Bureau of Economic Analysis and HKMA staff estimate

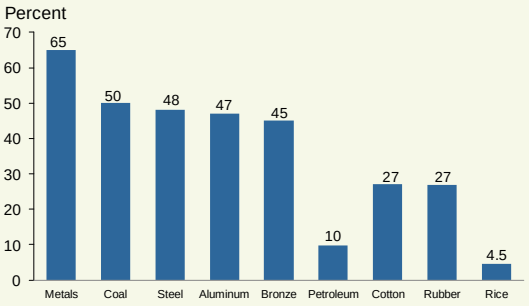
Meanwhile, China has increased its share of spending on domestically produced goods including raw materials, intermediate goods, and consumer goods in response to the country's ongoing economic reforms which promotes the development of domestic industrial clusters. The result of the reform policies is reflected by the increase in the ratio of goods produced for domestic consumption to total production in China across all goods categories **(Chart 4)**. Moreover, since 2011, the reforms have shifted the growth driver for the Chinese economy from an investment-led growth model to a consumption-led one. This has reduced China's reliance on commodity imports, which constitute the majority of imports from ASEAN-5. Indeed, China is the world's largest importer of commodities, with its share making up 40-60 percent of total global imports. Imported commodities are mostly used for investment purposes, i.e. steel, coal, aluminum and copper **(Chart 5)**.

Chart 4 Ratio of domestic consumption and exports to total production in China



Rationale: The figures show the ratio of domestic consumption to total production
Source: NBS

Chart 5 Ratio of China's commodity imports to total global imports



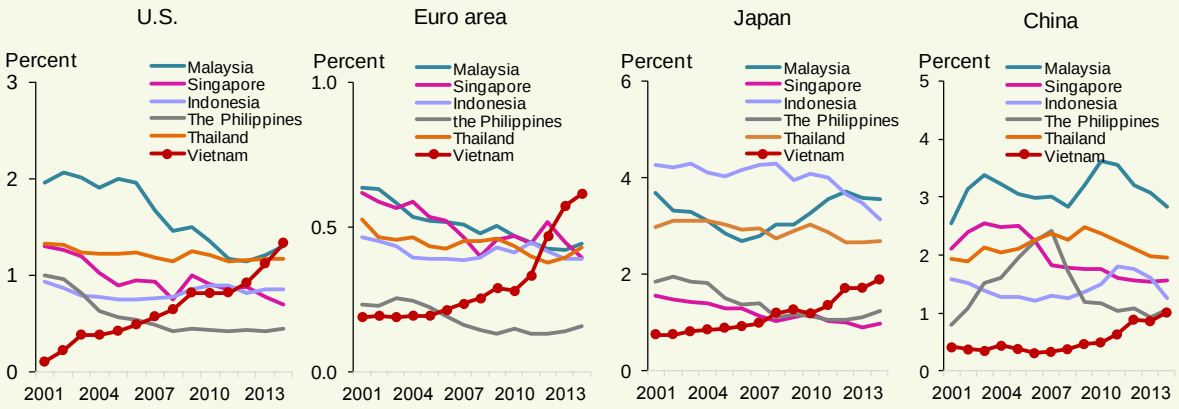
Source: IMF's article titled "China's Impact on World Commodity Markets"

(1.3) The development of industrial clusters in the ASEAN-5 countries. Foreign direct investment (FDI) in emerging markets over the past 10 years, which had gone into the development of upstream, midstream and downstream industries, have brought about a decline in the exports of intermediate goods by Asian countries. (For more detailed analysis, please refer to the article *"The Shifting Global Trade Structure"* in the June 2014 Monetary Policy Report.)

(2) A decline in the share of ASEAN-5 exports in the global market, dropping from 6.1 percent of total global exports in 2011 to 5.9 percent in 2014. By contrast, the global market share of exports from Emerging and Developing Europe[2/] grew from 3.1 percent to 3.4 percent, while the global market share of exports from CLMV countries[3/] increased from 0.6 percent to 1.1 percent during the same period. The decline in market share of exports of ASEAN-5 countries could be attributed to the following factors:

(2.1) ASEAN-5 countries' declining competitiveness against countries with lower production costs, especially labor costs such as Vietnam. This is reflected by the continued increase in Vietnam's share in total imports to the U.S., the euro area, Japan and China **(Chart 6)**. This is due partly to the relocation of the production base for electronics and electrical appliances from ASEAN-5 countries to Vietnam to take advantage of the lower labor costs.[4/]

Chart 6 Ratio of imports from ASEAN-5 and Vietnam to the total imports of the U.S., the euro area, Japan and China



Source: Trademap

(2.2) Some of the ASEAN-5 countries' failure to adapt to changing global consumer preferences, such as the shift in demands away from computers to smart phones and tablets. ASEAN-5's inability to adjust in a timely manner to build new productive capacity for smart phones and tablet-related products caused their exports to decline.

Based on the above factors, even if the global economy expands more robustly in the near future, the exports of ASEAN-5 countries may not benefit from global growth to the same degree as they did in the past. This further emphasizes the need for the ASEAN-5 economies to upgrade their production technology and adapt their production capability to meet the rapidly

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[2/] Emerging and Developing Europe consists of Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Hungary, Macedonia, Montenegro, Poland, Romania, Serbia and Turkey.

[3/] CLMV consists of Cambodia, Lao PDR, Myanmar and Vietnam.

[4/] Vietnam's average monthly income stands at approximately 212 USD, while Thailand's average monthly income is approximately 408 USD.

changing consumer preferences. Such adjustment would improve ASEAN-5's competitiveness and help maintain its market share in global exports, thus allowing the group to maximize the benefits from global economic recovery. It is, however, worth noting that some ASEAN-5 countries have already implemented economic reforms in response to the changing global trade structure. Singapore places a strong emphasis on the exports of services, which yield higher value-added than the exports of goods. Singapore is also striving to become the region's supply chain management hub. Malaysia has implemented its Economic Transformation Program (ETP) since 2010, with a goal to upgrade its production capability, boosting export values, and be more responsive to the rapidly changing global demand. Likewise, Thailand should place priorities on upgrading country's exports structure, in order to enhance its production capabilities and competitiveness in the global market.

Recent Economic Developments

2. Recent Economic Developments

The global economy expanded at a slower pace than previously projected mainly as a result of weaker-than-expected US economic growth in 2015 Q1, and economic slowdown in China and other Asian economies (excluding Japan) owing to sluggish exports. Going forward, the global economy is likely to face higher risks from the slowdown in the Chinese economy and uncertainty over Greece's financial situation.

The Thai economy continued to recover albeit at a slower pace, owing to the ongoing decline in exports since the beginning of the year. Weak exports adversely impacted employment, income and dented private sector confidence, prompting households to be more cautious regarding their spending. As a result, businesses delayed plans for capacity-expansion investment. Nonetheless, robust tourism growth and improving budget disbursements for public investment assumed greater roles in driving economic growth.

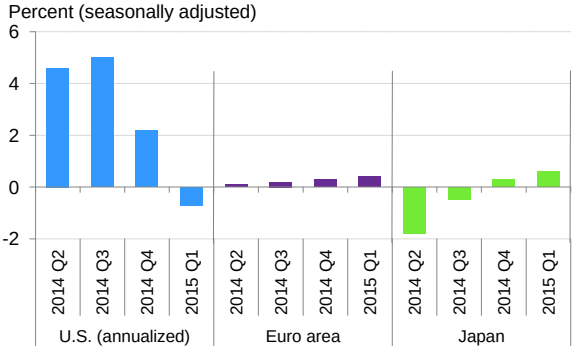
Inflationary pressures were generally weaker than assessed, due largely to cost factors. Headline inflation remained negative, but should gradually edge up in the latter half of the year. The MPC assessed the deflation risk in Thailand to remain low.

2.1 The global economy

The global economy expanded at a slower pace than assessed in the last monetary policy report due to slower growth in the U.S., Chinese and Asian economies. Meanwhile, the euro area economies exhibited clearer signs of recovery.

Advanced economies still posted positive growth despite a temporary slowdown in the U.S. **(Chart 2.1)**. In 2015 Q1, the **U.S. economy** contracted by 0.7 percent (qoq saar, second estimate), following 2.2 percent growth registered in the previous quarter. Economic activities were disrupted by temporary factors, such as the

Chart 2.1 GDP growth of G3 economies
(change from the previous quarter)



Source: Bureau of Economic Analysis, Eurostat, Cabinet Office of Japan

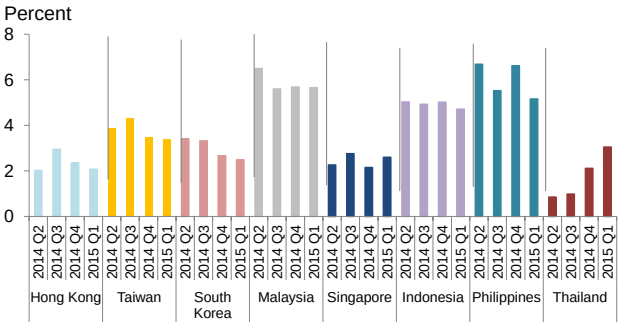
severe winter weather and dock workers protests at the West Coast ports. Moreover, stronger U.S. dollar increasingly weighed down net exports. Investment in the energy sector also declined by falling oil prices.

In contrast, **the euro area economy**[1/] expanded by 0.4 percent (qoq sa) in 2015 Q1, up from 0.3 percent recorded in the previous quarter as Spain, France and Italy posted higher growth. Meanwhile, the German economy grew at a decelerated pace. Overall growth pickup in the euro area economy was owing to higher private consumption, which benefited from low oil prices and more accommodative monetary conditions as well as improved exports in line with the weaker euro.

The Japanese economy continued to gradually recover, posting a positive growth of 0.6 percent (qoq sa) in 2015 Q1, up from 0.4 percent recorded in the previous quarter. Growth was driven mainly from higher inventory build-up, while consumption and private investment growth remained subdued. Exports continued to edge up in line with demands from major trading partners, particularly the U.S.

The Chinese and other Asian economies (excluding Japan) grew at a slower pace than expected **(Chart 2.2)**. In 2015 Q1, **the Chinese economy** expanded by 7.0 percent (yoy), down from 7.3 percent registered in the previous

Chart 2.2 GDP growth of G3 Asian economies
(change from the same quarter last year)



Source: CEIC

[1/] The euro area economy consists of 18 countries that share the euro as an official currency. As of 2013, Germany, France, Italy, and Spain contributed to 28, 21, 17, and 11 percent of the economy respectively, while Greece, Ireland, and Portuguese together accounted for 6 percent. Lithuania joined the group at the beginning of 2015, contributing to 0.4 percent of the 19-country currency union.

quarter, mainly due to moderating investment. Based on recent economic indicators at the beginning of 2015 Q2, investment, consumption and exports growth looked set to decelerate and fall below the targets set by the Chinese government. The government therefore introduced more expansionary monetary and fiscal measures to boost growth, such as the policy interest rate reduction on May 10, 2015, and efforts to expedite infrastructure investments by local governments.

The other Asian economies (excluding Japan) exhibited slower growth in 2015 Q1, owing primarily to weaker exports. In particular, exports to China and regional economies continued to decline, on the back of weak demands from trading partners and changes in global trade structure (*Article in Box 1: The Changing Global Trade Structure and Implications for Exports of ASEAN-5 Countries*). Preliminary economic data in 2015 Q2 indicated that the continued weakness in export growth has had adverse impacts on the real economy. Manufacturing production has continued to slow since the beginning of 2015, consistent with the Purchasing Managers' Index (PMI) being below 50 in almost every country in the region. Domestic demand in Asian economies (excluding China and Japan) began to show signs of moderating, with falling investment in machinery and equipment reflected by the decline in imports of machinery and equipment in several countries. Despite sharply lower oil prices, private consumption began to fall because declining manufacturing employment and weaker consumer confidence led the private sector to be more cautious with spending.

Looking ahead, the Committee judged that the global economy would continue to recover, albeit at a more modest pace. Downside risks to growth are now more than previously assessed, especially

from the Chinese economic slowdown. Additional risks are from the uncertainty over finding a resolution to Greece's financial rescue plan, and divergent monetary policies of major advanced economies, which could lead to greater volatility in the financial markets.

The U.S. economy will gain momentum from improving economic fundamentals, especially from the continued recovery of the labor market where employment and wages are picking up. The private sector balance sheet will also strengthen as a result of continued deleveraging. Nonetheless, further risks remain from the sustained strength of the U.S. dollar and greater financial market volatility stemming from the expectation of the Fed's interest rate hike.

The euro area economy are projected to recover gradually from the ECB's continuation of easing monetary policy. Going forward, however, economic activities and confidence in the euro area could weaken if the Greek government and its creditors fail to reach an agreement on debt bailout terms and structural reforms.

The Japanese economy is expected to expand at a gradual pace. The conclusion of the negotiation on annual wage rise could lead to a significant increase in wages in mid-2015, which should provide a boost to consumption. Export growth should benefit from continued yen weakness, but could be curtailed by the slowdown in China and the rest of Asia. However, the Japanese economic growth in the long run is largely dependent on the outcome of structural reforms.

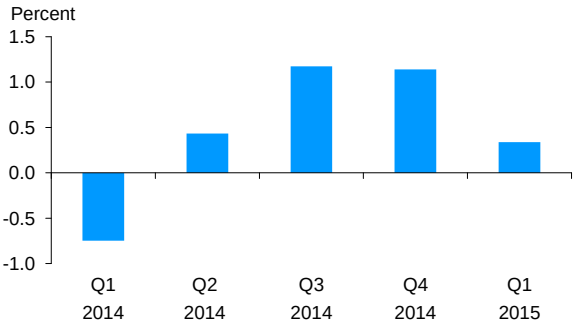
The Chinese and other Asian economies are likely to grow at a slower pace, due mainly to weaker exports. In the periods ahead, exports from China and other Asian countries (excluding Japan)

are projected to edge up slowly. However, exports growth is expected to be lower than the rate recorded in the years preceding the global economic crisis, because of the ongoing slowdown in China and the shifts in global trade structure.

2.2 The domestic economy

The pace of Thailand's economic recovery slowed in 2015 Q1 **(Chart 2.3).** The decline in exports since the start of the year was a result of sluggish global recovery and structural changes in global trade that altered the contribution of global recovery to Thai economic growth. Weaker exports lowered employment, income and private sector confidence, prompting households to restrain spending. Likewise, the business sector cut production and postponed investment that would upgrade production capacities. In contrast to these setbacks, the tourism sector continued to expand and disbursements of government investment budget became important drivers of growth. The policy rate reduction helped lower financial burden to some degree, but its effects on domestic demand were not evident yet.

Chart 2.3 GDP growth[1/]
(seasonally adjusted, percentage change from last quarter)

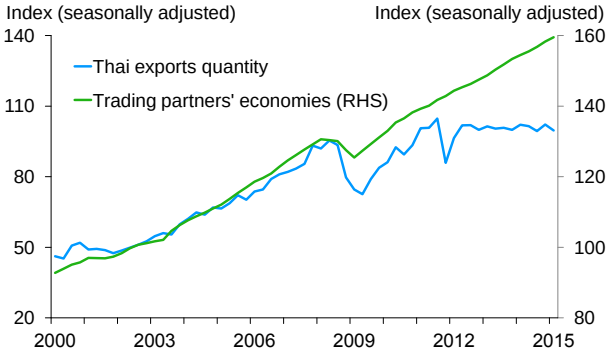


Note: [1/]Calculations based on chain volume measure (CVM)
Source: Office of the National Economic and Social Development Board

Exports fell in quantity and price terms

Exports contracted from the previous quarter, both in terms of quantity and price, due to structural and cyclical factors. Structural problems included the following: (1) Structural changes in global trade prompted major trading partners to rely less on imports. As a result, Thai export quantity benefited less from the recovery in trading partners' economies than in the past **(Chart 2.4)**. (2) Thailand's international competitiveness declined as a result of higher

Chart 2.4 Thai exports and trading partners' economies

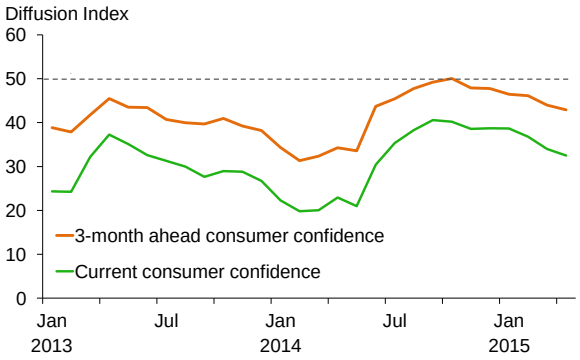


Source: Trademap, Ministry of Commerce, calculations by Bank of Thailand

wages, constraints in production technology, and termination of GSP privileges for Thai exports to the euro area (*Article in Box: Structural Problems in Thai Exports Sector*). Cyclical factors were: (1) The slowdown in trading partners' economies, especially China, hurt exports of ASEAN countries. (2) Prices of several export products, such as commodities, plastics and chemicals, remained low in line with crude oil price.

Going forward, the Thai exports are constrained by both the slowdown in trading partners' economies and structural issues. Exporters need time to explore new markets and develop higher value-added products. However, in the short term, the recent baht depreciation should help increase liquidity for Thai exporters somewhat via higher revenue in baht term.

Chart 2.5 Consumer Confidence Index

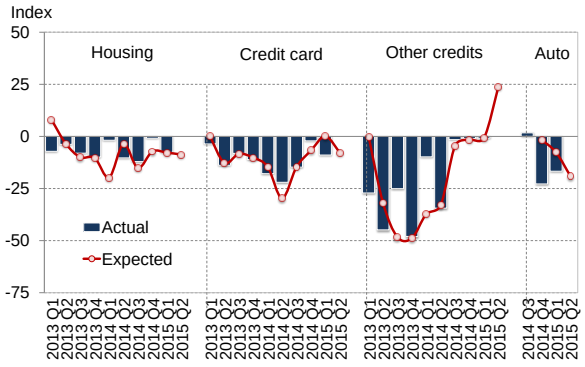


Source: Ministry of Commerce, calculations by Bank of Thailand.

Households remain cautious with regards to spending

Private spending recovery remained sluggish as non-farm incomes fell (*Article in Box: Changes in Household Incomes and Implications for Economic Recovery*). Other factors, i.e. falling farm incomes and private sector confidence, continued to weigh on private consumption **(Chart 2.5)**. Household debt levels were persistently high, while financial institutions remained strict with lending standards to households as the quality of consumer loans worsened **(Chart 2.6)**. In the periods ahead, consumption is likely to pick up slowly amidst these existing constraints. Although employment in 2015 Q2 edged up slightly from the past quarter, the increase was mostly in the sectors with relatively low wage per hour. Therefore, consumers continued to remain cautious with their spending amidst concerns over future income prospect. Nonetheless, the decline in

Chart 2.6 Lending standards to household sector



Note: Index > 0 indicates less strict standards; Index = 0 indicates stable standards; Index < 0 indicates stricter standards

Source: Survey of credit conditions, Bank of Thailand.

commercial banks' lending rates following the policy rate reduction should alleviate some of the debt burden for households. In addition, some public financial institutions are also preparing to relax their lending standards in 2015 Q2 to provide support to private consumption, in line with government policy.

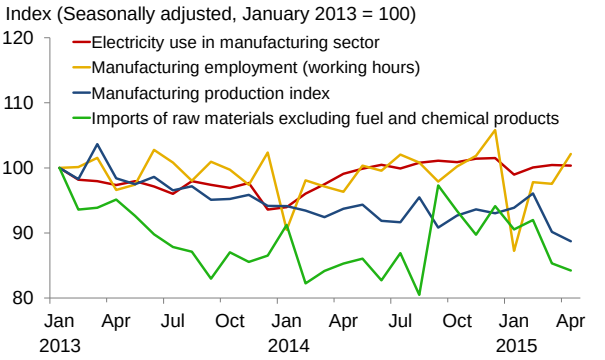
Weak domestic and external demand prompted firms to cut back production and delay investment plans for capacity expansion.

Manufacturing production edged down from the previous quarter from the decline in manufacturing exports (Chart 2.7). Hard Disk Drive (HDD) production fell significantly because more consumers switched to advanced computer components made of solid-state drive (SSD). Meanwhile, some HDD factories also shut down for maintenance. Electronics production contracted amidst falling domestic and external demand, compounded by the relocation of production base of major TV manufacturers to other countries in the region. Nevertheless, production of some manufacturing items expanded. For example, automobile production for exports grew, especially in eco-cars and pick-up trucks destined for the Australian and European markets. In addition, integrated circuits (IC) manufacturing was still in an expansionary phase **(Chart 2.8)**.

Going forward, the recovery of the manufacturing sector will hinge mainly on the recovery of domestic and external demand, along with the ability to upgrade Thailand's competitiveness.

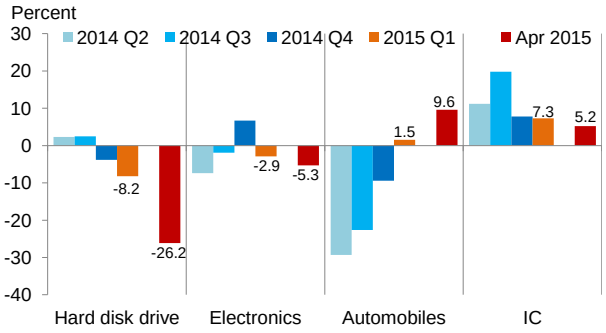
Private investment expanded at a slower pace amidst weak domestic and external demand. The business sector still had ample spare production capacity, particularly exports manufacturers whose capacity utilization was lower than other sectors

Chart 2.7 Manufacturing production conditions



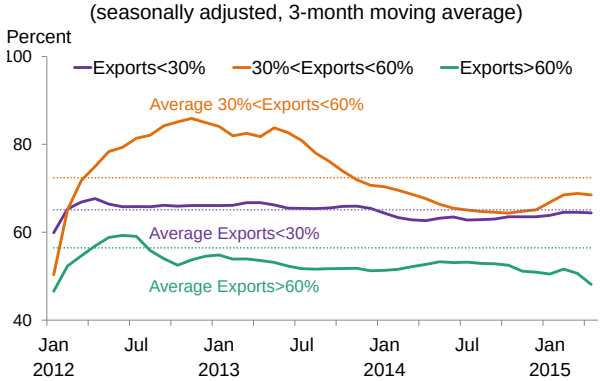
Source: Office of Industrial Economics, Ministry of Industry, National Statistical Office, The Customs Department, The Thai Automotive Industry Association, and the Bank of Thailand

Chart 2.8 Manufacturing production by sectors
(compared with the same period last year)



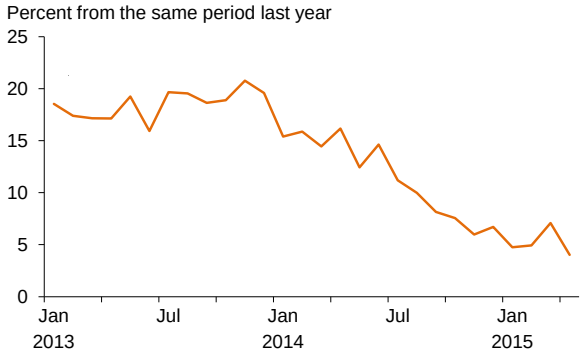
Source: Office of Industrial Economics, Ministry of Industry

Chart 2.9 Production capacity by exports
(seasonally adjusted, 3-month moving average)



Source: Office of Industrial Economics, Ministry of Industry, calculation by Bank of Thailand

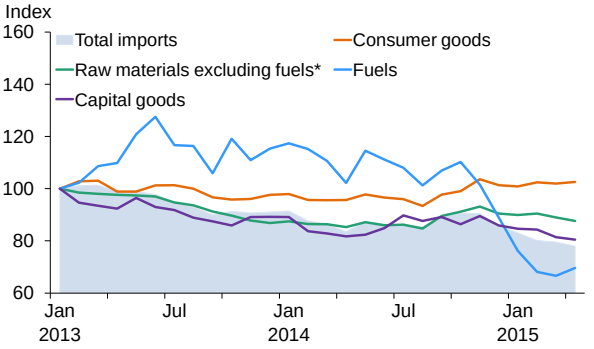
Chart 2.10 Commercial banks' credit for investment in fixed assets



Source: Ministry of Commerce, calculations by Bank of Thailand

Chart 2.11 Imports index, by product groups

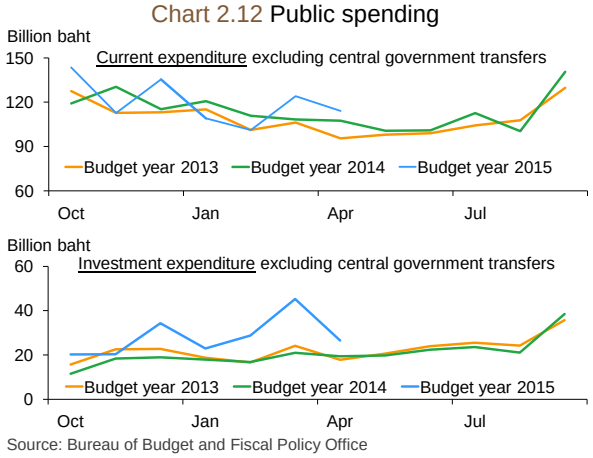
(seasonally adjusted, 3-month moving average, Jan 2013 = 100)



Note: *Auto parts included

Source: Customs Department (data processed by Bank of Thailand)

Chart 2.12 Public spending



Source: Bureau of Budget and Fiscal Policy Office

(Chart 2.9). The private sector also held back on new investment, awaiting greater clarity on economic recovery and government plans on infrastructure investment. This is consistent with declining private sector confidence and private credit **(Chart 2.10)**.

Imports dropped in line with subdued production and investment

Imports edged down from the previous quarter (Chart 2.11). Imports of raw materials, excluding fuels and chemicals, contracted in line with declining manufacturing production. Imports of capital goods continued to shrink, consistent with subdued investments. Fuel imports remained at very low level due to weak crude oil prices. Imports of consumption goods edged up slightly because the rise in the number of foreign tourists led to higher imports of some consumer products, such as cosmetics and watches.

Tourism growth and public spending, especially investment expenditure, became important driving forces for the economy.

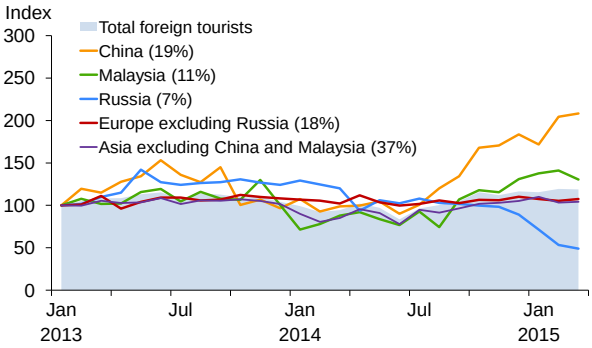
Public spending, particularly investment expenditure, rose from the past quarter (Chart 2.12). Broadly consistent with the government's policy, public agencies were urged to expedite government procurement before the enactment of the Annual Budget Expenditure Act and to enter into contracts for public projects before December 2014. Consequently, disbursements picked up pace in February and March 2015. However, public investment spending moderated slightly in April because it was not the period scheduled for the project handover and budget disbursement of investment projects.

Going forward, public spending will continue to play a key supportive role for economic growth. The government already approved the second phase of economic stimulus comprising road transport infrastructure and water management system projects. Quick disbursements for these investment plans are likely because they consist mostly of small and medium projects with most completion due within one year. Local government spending is expected to pick up slightly in the second half of the budget year, due to the expedited signing of contracts for government projects and the tendency that disbursements of subsidies are normally back-loaded. However, several government megaprojects are still in their initial phase and will take some time to materialize. Examples of ongoing projects are rail system development, inter-city motorway construction, and highway development linking the East-West Corridor.

Tourism posted a solid growth owing to the growing number of Chinese and Malaysian tourists, which helped offset the drop of Russian tourists attributed to Russia's economic woes and sharp ruble weakness **(Chart 2.13).**

Looking ahead, tourism is projected to continue growing despite a downside risk concerning Thailand's international aviation safety standards.[2/] This concern is unlikely to significantly impact the number of foreign tourists because this issue mainly centers on charter flights. Foreign tourists will be able to travel by scheduled flights

Chart 2.13 Number of tourists by origins
(Indices, January 2013 = 100)



Note: Parentheses ( ) indicate shares of total foreign tourists in 2014
Source: Department of Tourism

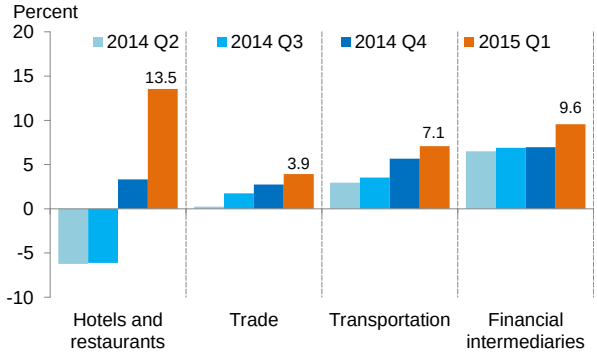
[2/] In March 2015, the International Civil Aviation Organization (ICAO) announced that a safety oversight audit conducted in Thailand found "Significant Safety Concerns" (SSC) in two areas, namely (1) air operator certification procedures and (2) transport of dangerous goods.

operated by domestic airlines or any other flights operated by foreign airlines. Moreover, the tourism sector is likely to benefit from the lifting of martial law on April 1, 2015, as some foreign governments have removed warnings against travel to Thailand and as a result, tourists are now able to purchase travel insurance.

Service sector posted a solid growth on robust tourism industry

Tourism-related service sectors saw a strong growth. Hotels, restaurants and transportation, especially air transport, posted a solid growth. These businesses benefited from the rise in the number of foreign tourists and added support to the retail sector through tourists' spending. Other service sectors also exhibited an elevated growth. The telecommunications sector grew from increased demand for data services. Financial intermediaries saw the expansion of services by other depository institutions, as well as by insurance and life insurance companies **(Chart 2.14)**.

Chart 2.14 Growth of the service sectors



Source: Office of the National Economic and Social Development Board

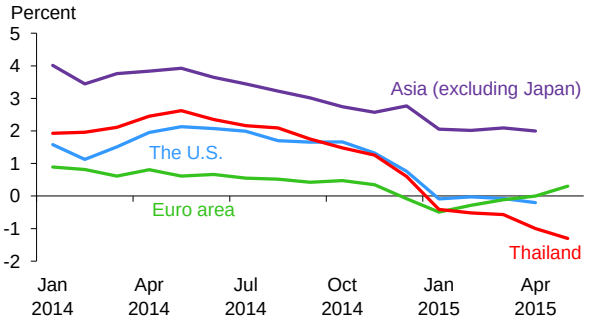
2.3 Production cost and price conditions

Inflationary pressure was weaker than previously assessed. Headline inflation edged down and is expected to remain in negative territory for a longer period, mostly due to depressed energy and fresh food costs. However, headline inflation is expected to pick up in the second half of the year, bottoming out in Q2 before gradually rising in Q3 and turning positive in 2015 Q4. Although core inflation edged down mainly from food costs, it is expected to remain

positive throughout the forecast period. The Committee considered the overall deflation risk to be low.

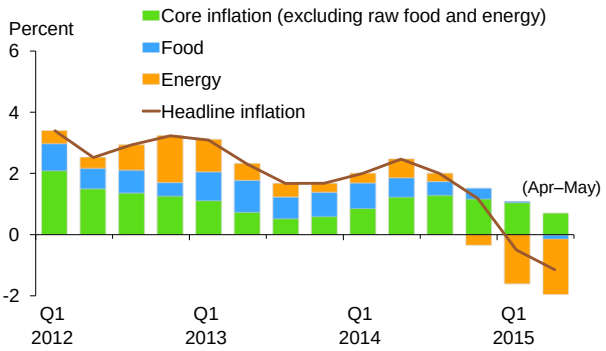
Recently, inflation readings in several countries including those in Asia began to stabilize, consistent with global crude oil prices and clearer signs of economic recovery especially in the euro area **(Chart 2.15)**. Thailand's general price level continued to decline, with headline inflation recording an average of negative 1.16 percent for the first two months of 2015 Q2 **(Chart 2.16)**, mainly due to cost factors. Despite a pickup in global oil prices and recent baht depreciation, domestic retail oil prices remained stable at low levels following the government's decision to lower the contribution to the Oil Fund.[3/] Moreover, declining headline inflation also reflected the reduction in the fuel adjustment charge (FT) during May-August 2015 by 0.0935 baht per unit, in line with lower fuel costs. Falling raw food prices from an oversupply of meat and eggs, combined with weaker-than-expected demand recovery, also contributed to the persistence of negative headline inflation. Negative headline inflation was previously projected to remain until 2015 Q2, but is now expected to last well into 2015 Q3. Meanwhile, core inflation edged down to 0.94 percent in May 2015, mostly from lower raw food costs that contributed to subdued food prices. Prices of other goods and services rose from the same period last year, but remained mostly stable relative to 2015 Q1, in line with weak domestic demand **(Chart 2.17).** This was consistent with underlying

Chart 2.15 Domestic and foreign headline inflation



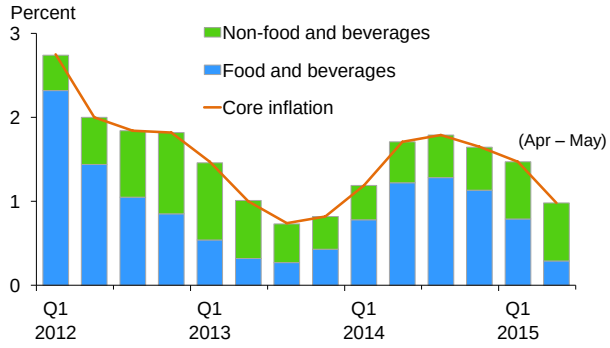
Notes: Foreign headline inflation calculated from unweighted averages of individual countries inflation
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, CEIC, and calculations by Bank of Thailand

Chart 2.16 Contributions to headline inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, and calculations by Bank of Thailand

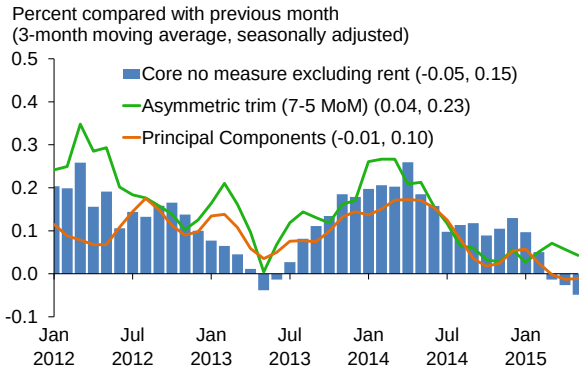
Chart 2.17 Contributions to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, and calculations by Bank of Thailand

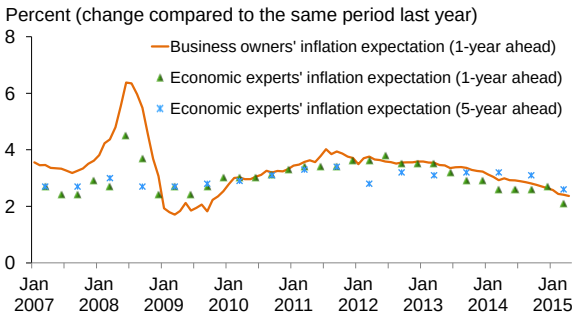
[3/] The government cut contribution to the Oil Fund 4 times during March–May 2015. Consequently, the prices of benzene, namely gasohol 95, E20 and gasohol 91, and diesel fell by 3.00, 2.80, 2.10 and 2.30 baht per liter respectively.

Chart 2.18 Underlying Inflation



Notes: Inside parentheses, figures on the left indicate May 2015 %MoM_SA (3mma), while those on the right show 2010-2014 average.

Chart 2.19 Public inflation expectations



Note: Business owners' inflation expectation surveyed by Bank of Thailand
Economic experts' inflation expectation surveyed by Consensus Economics
Source: Business Sentiment Index Survey by Bank of Thailand and Asia Pacific Consensus Forecast

inflation,[4/] which indicated a general decline in inflationary pressure and limitation on the business sector to raise prices **(Chart 2.18)**.

Nonetheless, going forward headline inflation is expected to pick up mainly from cost factors. In particular, the high base effect of oil prices during the first half of 2015 will gradually diminish. Moreover, global crude oil prices are expected to increase, while raw food prices are likely to edge up when the problem of an oversupply of agricultural products begins to subside. At the same time, demand-pull inflationary pressure is projected to increase slowly in tandem with economic recovery. The MPC thus judged that headline inflation will bottom out in 2015 Q2, before picking up in Q3 and turning positive in the last quarter of this year. Core inflation remained positive, but is likely to trend down until early 2016 due to its high base in the previous year and weak demand-side pressure.

Although headline inflation is expected to remain negative for longer than previously assessed, the Committee considered deflation risk to be low because private consumption continued to expand, while negative headline inflation resulted mainly from falling energy and raw food costs. Prices of most goods and services also remained stable or rose. In addition, short- and medium-term inflation expectations, while declining slightly, were still close to the inflation target **(Chart 2.19)**. This suggests that the possibility of a downward spiral that could lead to deflation remains low. Likewise, deflation indicators pointed to low deflation risk over the coming months. While slower- or weaker-than-expected recovery could

[4/] Please refer to FAQ issue 93 "Indicators for inflation trend" at www.bot.or.th.

give rise to higher deflation risk in the future, the overall risk level is deemed not to be a cause for concern *(Article in Box: Assessing Deflation Risks in Thailand)*.

Table 2.1 Quarterly inflation

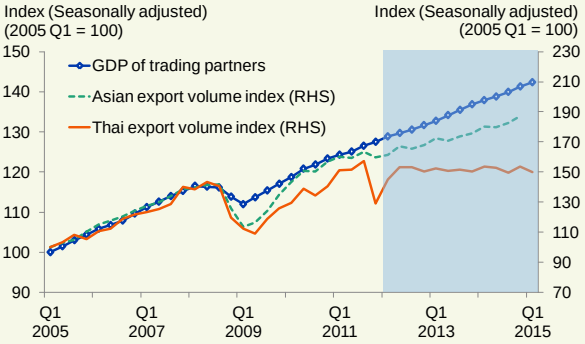
Unit: Percent	2013	2014				2015	
		Q1	Q2	Q3	Q4	Q1	Apr-May
Percentage change from previous year (%Δyoy)							
- Headline Consumer Price Index	2.18	2.00	2.47	2.00	1.11	-0.50	-1.16
• Core Consumer Price Index	1.00	1.19	1.70	1.79	1.65	1.47	0.98
• Raw food	5.54	5.34	3.91	2.78	1.88	0.30	-0.54
• Energy	4.79	2.55	5.13	2.20	-3.11	-13.0	-14.3
Percentage change from previous quarter (%Δqoq_sa)							
- Headline Consumer Price Index	-	0.8	0.5	0.0	-0.2	-0.8	-
• Core Consumer Price Index	-	0.5	0.6	0.3	0.2	0.3	-
• Raw food	-	1.9	0.1	-0.6	0.5	0.3	-
• Energy	-	0.5	0.6	-0.7	-3.5	-9.7	-

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, and seasonally adjusted quarter-on-quarter percentage change calculations by Bank of Thailand.

Structural problems within the Thai exports sector

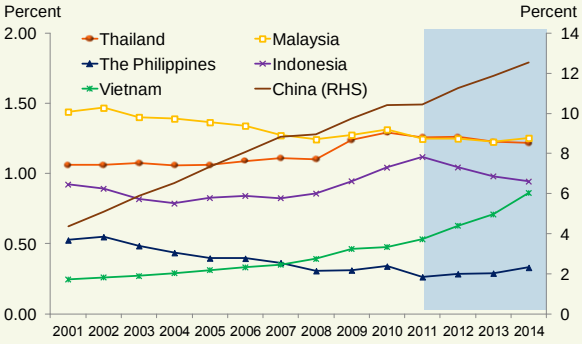
The slowdown of Thai exports over the past three years has been in part attributed to the global trade slowdown, an external factor. It also, however, stemmed from **a number of domestic structural problems within the Thai exports sector that undermined the country's competitiveness**. The Thai export recovery was therefore more sluggish than that of other countries in the region **(Chart 1)**. Moreover, the share of Thai exports in the global market has constantly declined **(Chart 2)**, while Chinese, Vietnamese and the Philippines exports continue to gain market shares. In case of Malaysian and Indonesian, though their market shares for global exports have fallen, the decline is due mainly to lower prices of energy commodities, which are their key exports

Chart 1 GDP of trading partners and comparison between Asian and Thai exports



Source: WTO and Bank of Thailand

Chart 2 Share of developing Asian countries in the global export value



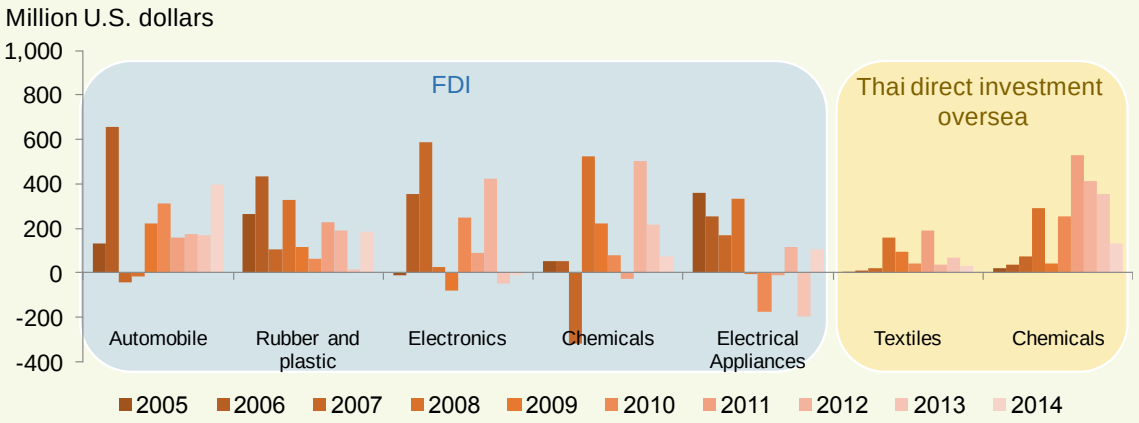
Source: Trademap

Domestic structural problems could result from the following factors:

1. Thai export quantity expanded at a lower rate than the past average because of a persistent lack of new investment to expand existing capacities. Since 2012, Foreign Direct Investment (FDI) flows have slowed in several industries **(Chart 3)**. By contrast, Vietnam, the Philippines, and Indonesia have managed to consistently attract FDI inflows from large Multi-National Corporations (MNCs), due largely to lower labor costs, large labor forces, vast domestic markets and abundant natural resources.

At the same time, outward direct investment by Thai businesses rose steadily. The steady increase in Thai investment abroad in capital-intensive industries, such as petrochemicals and chemicals, was motivated by the environmental concerns of these industries in Thailand. Labor-intensive industries, such as textiles and apparels, have also relocated to neighboring countries like Vietnam and Cambodia, to take advantage of lower wages and greater tax incentives under Generalized Scheme of Preferences (GSP) and bilateral Free Trade Agreements (FTAs). Thai exports of these products, therefore, either stagnated or declined steadily.

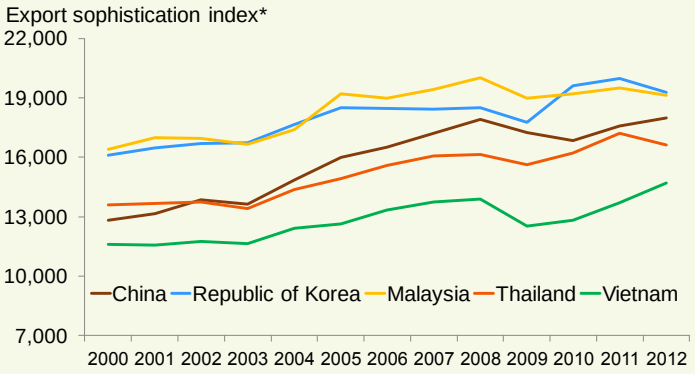
Chart 3 Thailand's FDI inflows and Thai direct investment overseas*



Note: *Only equity investment and subsidiary loans
Source: Bank of Thailand

2. Our export quality failed to match the improvements made by our regional competitors **(Chart 4)**. Thailand's Export Sophistication Index declined in recent years, while those of China and Vietnam posted continued gains, pointing to Thailand's diminishing quality competitiveness and greater reliance on unsophisticated exports. Quantity of exports, particularly for farm and processed food products, is also sensitive to labor costs, tax rates and the exchange rate movements. Moreover, in the electronics sector, hard disk drive exports suffer from steadily lower global demand. Despite the increase in export quantity of eco-cars, the value of total exports might not improve significantly because these type of cars command low price per unit.

Chart 4 Export sophistication index



Note: *The Export Sophistication Index measures the quality of the structure of a country's exports. A high index value reflects an export structure with a concentration of goods generally produced by high income countries.
Source: IMF

Stagnant quality of Thai exports is likely caused by the following key factors:

◆ Thailand's progress on upgrading its research and development (R&D) capabilities lags behind neighboring countries. Thailand has, for a long time, neglected the importance of R&D, unlike other regional countries where there have been sustained efforts, led by the government, to develop research infrastructure. The Global Competitiveness Report 2014-2015 clearly highlights this problem **(Table 1)**. The quality of institutions and human resource readiness in R&D in Thailand has been falling behind countries in the region. Likewise, Thailand still lacks the necessary basic infrastructure that would create an environment conducive for research,

Table 1 Innovation components of the Global Competitiveness Index (Rank / 144 countries)

	Company spending on R&D	Quality of scientific research institutions	Availability of scientists and engineers	Patent Cooperation Treaty (PCT) application	Government procurement of advanced technology products
Thailand	56	61	54	104	114
Malaysia	9	20	9	25	3
Indonesia	24	41	31	43	13
China	23	39	43	53	10

Source: World Economic Forum

namely effective protection of intellectual property rights as well as clear and concrete government policies to promote innovation.

◆ The quality of human capital in Thailand is lower than many other countries in the region. According to the 2015 human capital index, Thailand was ranked 57th out of a total of 124 countries, well below neighboring countries like the Philippines and Malaysia. Despite a higher rank than Vietnam, China and Indonesia, Thailand's score does not differ significantly from them. These countries also have larger populations than Thailand. Thailand's score highlights the inadequate development of education quality and labor skills. These human capital factors are vital for both improving export sophistication and attracting FDI inflows into high value-added industries **(Table 2)**.

Table 2 Human Capital Index 2015*

Rank	Country	Score
5	Japan	82.74
24	Singapore	78.15
30	Republic of Korea	76.84
46	The Philippines	71.24
52	Malaysia	70.24
57	Thailand	68.78
59	Vietnam	68.48
64	China	67.47
69	Indonesia	66.99

Note: *Measures the quality of education and employment among 5 age groups
Source: World Economic Forum (Human Capital Report 2015)

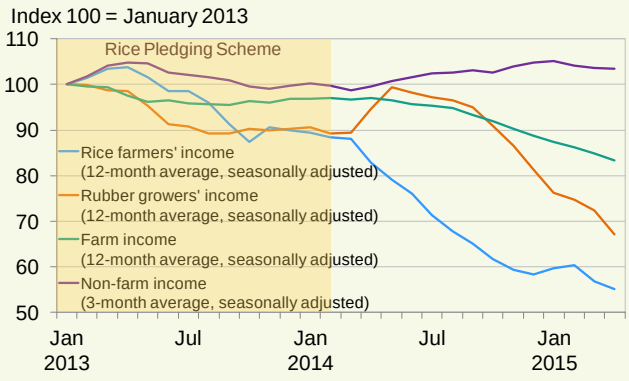
These structural problems have been accumulating for quite some time, but the consequence has become much more visible today. In addition to these domestic issues, structural changes in global trade have weakened the links between the volume of international trade and global growth. These internal and external constraints severely limit the ability of Thai exports to enjoy the high growth rates like in the past. While structural changes in global trade are an external factor, beyond our control, the domestic structural constraints can be addressed, which will undoubtedly take some time. **Rectifying these problems requires close multi-sectoral cooperation, guided by a clear and coordinated reform plan.** The academic sector should play a role in conducting research and formulating clear approaches to solutions, while the government should support labor skills development and encourage investment that would enhance production capabilities and improve product quality. In addition, the private sector should be proactive in upgrading their competitiveness and product quality in order to avoid price competition in the long term. Unless these internal structural problems are firmly addressed, it will be difficult for the Thai exports sector to resume its usual role as the main driver of growth, which will hurt Thailand's long-term economic potential.

Changes in household income and implications for economic recovery

Household income is an important determinant of private consumption, a major driver of the Thai economy. In recent years, however, household income was hit by the global economic slowdown, causing low agricultural prices and weak demand for Thai exports. Farm income, therefore, remained subdued in tandem with weak agricultural prices. Moreover, employment in the manufacturing sector recovered only slowly, further depressing household non-farm income. Weaker fundamentals of both farm and non-farm sectors played major role in impeding the recovery of private consumption in recent years.

Farm income[1/] remained low. Farm income has continuously declined over the past two years **(Chart 1)**, due mostly to dwindling income from rice and rubber sales, which together accounted for 45 percent of total farm income. Domestic rice prices plunged following the end of the rice pledging scheme in February 2014. Demand for rubber from large consumers, especially China and Malaysia, posted a significant drop, causing rubber prices to have fallen since June 2014. Moreover, drought and low levels of irrigation water in 2014 prohibited off-season rice farming in almost half of the cultivated areas, further depressing farm income. Going forward, the impacts of these drawbacks will be felt for some time, putting downward pressure on the purchasing power of farm households for the remaining of this year.

Chart 1 Farm and non-farm income



Source: Office of Agricultural Economics, National Statistical Office, and calculations by Bank of Thailand

Non-farm income[2/] started to edge down **(Chart 1)** in 2015 Q1, against the backdrop of falling exports. In contrast, the overall non-farm employment rebounded, in line with strong tourism growth and gradual recovery of domestic spending. However, most of recent non-farm employment took place in sectors with low to medium hourly wages **(Chart 2 and Chart 3)**, such as computers and electronics, food and beverages, hotels and restaurants, retail and land transport. While employment recovered only slowly in industries with high hourly wages, for example automobiles, hard disk drives and chemicals, as well as in service sectors, such as telecommunications and financial intermediaries. Therefore, improved

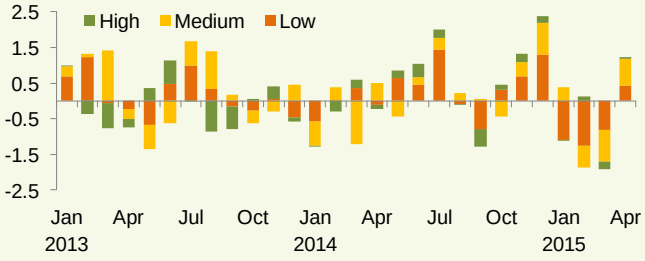
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[1/] The agricultural sector employs 12.7 million people (33 percent of total employed in 2014). The three most grown crops are rice, rubber and cassava, which account for 65, 25 and 10 percent of farm households respectively. Each farm household grows an average of 1.6 types of crops.

[2/] The non-farm sectors employ 25.3 million people (67 percent of total employed in 2014) across a number of sectors: 6.4 million in manufacturing, 6.2 million in trade, 2.6 million in hotels and restaurants, 2.3 million in construction, 1.3 million in transport and telecommunications, and 6.5 million in others.

Chart 2 Employment growth in manufacturing sector

Percentage change from previous month
(3-month moving average, seasonally adjusted)

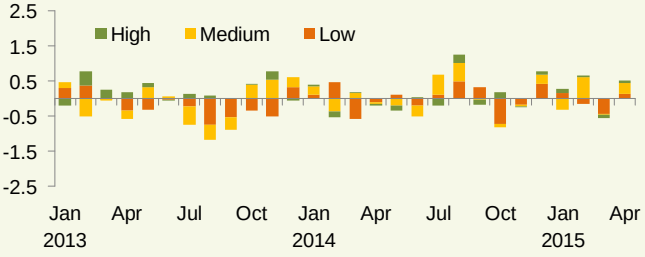


Note: Employment growth in manufacturing sector categorized by hourly wage
High = 76-100 Percentile, e.g. automobile industry
Medium = 34-75 Percentile, e.g. paper industry
Low = 0-33 Percentile, e.g. food and beverages
*The hourly wage for each industry is the average hourly wage during the Jan 2014 – April 2015 period.

Source: National Statistical Office and calculations by Bank of Thailand

Chart 3 Employment growth in service and construction sectors

Percentage change from previous month
(3-month moving average, seasonally adjusted)



Note: Employment growth in manufacturing sector categorized by hourly wage
High = 76-100 Percentile, e.g. financial intermediaries sector
Medium = 21-75 Percentile, e.g. retail
Low = 0-20 Percentile, e.g. restaurants
* The hourly wage for each industry is the average hourly wage during the Jan 2014 – April 2015 period.

Source: National Statistical Office and calculations by Bank of Thailand

conditions for employment have not materially boosted non-farm income. Over the coming months, constraints on exports recovery will continue to weigh on non-farm income and private consumption.

Looking ahead, the overall recovery of farm and non-farm income will be subject to these constraints, unable to provide a major boost for private consumption. Nevertheless, employment fundamentals generally remain sound, helping households retain some degree of spending power. Moreover, the migration of labor from farm to non-farm sectors, both manufacturing and services, leads to higher income for workers. Although these business sectors tend to pay low to medium wages, non-farm pay is still higher than farm income during this period of subdued agricultural prices, contributing to higher income and purchasing power for these workers. Furthermore, Thailand enjoys higher productivity gains due to the migration of labor from farm sector, characterized by low productivity, to manufacturing and service sectors with higher productivity. However, over the long term, the lifting of our labor income requires rising labor productivity, both in farm and non-farm sectors. This can be done through labor skill development and adoption of appropriate innovation in production.

Assessing the probability of deflation risks in Thailand

With headline inflation remaining negative and declining by an average of 0.8 percent for the first five months of 2015, there could be mounting concerns over the risks of deflation in the Thai economy. The MPC recognizes the importance of communicating with the public the definition, causes and impacts of deflation on the economy and financial stability—and the assessment of the probability of deflation risks in Thailand.

There are **narrow and broad definitions of deflation**.[1/] **Defined narrowly**, deflation refers to a sustained contraction of overall prices. However, this narrow definition is rather incomplete because it does not factor in the causes and consequences of deflation. On the one hand, deflation may arise from **supply-side factors**. For example, higher labor productivity, more advanced technology, and lower costs of inputs could translate into lower production costs and a general decline in prices, which are beneficial to the economy. On the other hand, deflation may be driven by **demand-side factors**. For example, in a prolonged recession, people may expect a fall in future prices and, therefore, delay consumption. This will discourage production and private investment, lowering employment, wages and income, and ultimately creating a deflationary spiral. Therefore, an assessment of deflation risks should make use of the broader perspectives that take into account the causes and consequences of deflation. **Broadly defined**, deflation arises when the price level is persistently falling across a broad-based range of goods and services, and when the central bank fails to anchor public inflation expectations. In this broader view, deflation often occurs during a period of a prolonged contraction of demand and employment, as was the case of the U.S. and the UK in the 1930s and Japan from the late 1990s to the early 2000s.

Taking this broader perspective, the Thai economy is not experiencing deflation. This is because the current negative headline inflation stems mostly from the sharp decline in global crude oil prices given the excess supply of oil. This development benefits the Thai economy, given the country's heavy dependence on imports of oil. Subdued oil prices also reduced costs for household and business sectors, while helping to support private consumption. In addition, the fall in prices is concentrated in energy and food categories from supply-side factors. Although the public's inflation expectations moderated, they remain close to the inflation target. More importantly, consumption continued to expand. Looking ahead into the second half of the year, the effects of a high base of oil prices will gradually diminish, while global oil prices and fresh food prices are expected to pick up. Taking into account these factors, the Committee projected **headline inflation to bottom out in 2015 Q2, before picking up and finally turning positive in 2015 Q4**.

[1/] For details on definition of deflation, refer to the article by European Central Bank (2014), Monthly Bulletin, June 2014.

Core inflation edged down in early 2015 but remains positive. The decline in core inflation stemmed from lower food prices, amidst an oversupply of meat and eggs, as well as, feeble demand. **If domestic demand turns out to be weaker than the current projection, it could put downward pressure on core inflation to drop further in the future.**

Apart from assessing the outlook for inflation, the Committee considers other factors that could elevate the risks of deflation as broadly defined. The BOT has developed the **Combined Deflation Risk Index (CDRI)** as a tool for evaluating the probability of deflation risks in Thailand. In developing the index, the BOT integrated the concepts for index construction used by the ECB and the IMF[2/]. The index calculation is based on different risk factors that could give rise to the kind of deflation which undermines economic and financial stability. There are a total of 15 variables divided into 5 categories as follows:

(1) Costs and prices including changes in price of goods and services, producer price index, real wages, and GDP deflator. The broadness and persistence of the decline of prices are taken into account.

(2) Inflation expectation is measured by the difference between current inflation and inflation expectation 12-month ahead. A significant fall in inflation expectation reflects the unanchoring of public inflation expectation, which encourage households to delay consumption in anticipation of lower prices, thereby reducing current business earnings.

(3) Excess capacity including the output gap and the average real GDP growth in the medium term relative to the long term. If there is large excess capacity, there will be little incentive to invest.

(4) Asset prices and exchange rate including changes in the stock exchange index and the real effective exchange rate (REER). A fall in asset prices suggests declining wealth, while a rise in the REER reflects lower competitiveness, affecting exports revenue.

(5) Financial factors including credit and money supply growth. Slow growth of credit and money supply point to sluggish economic activities.

Each variable is assessed against a criteria.[3/] A score of 1 is given to a variable that meets the criteria, and a score of 0 otherwise. The unweighted average[4/] is then multiplied by 100 to obtain an index with a possible value ranging from 0 to 100, with the value of 50 and

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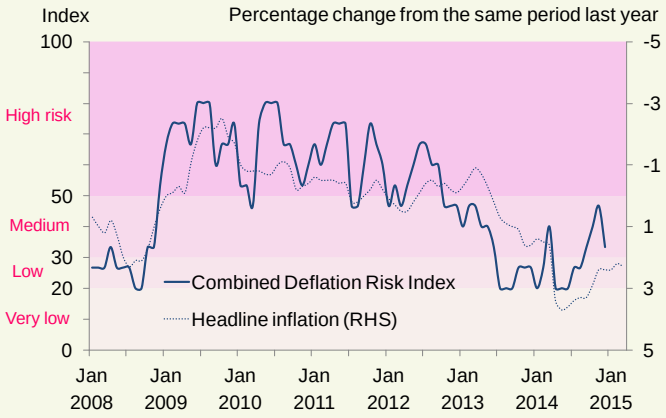
[2/] For more details on conceptual frameworks, see Norges Bank (2014) Deflation Indicator for the Euro Area, Economic Commentaries No.1/2014, and Decressin, J., and Douglas L. (2009) Gauging Risks for Deflation, IMF Staff Position Note. SPN/09/01.

[3/] Criteria used are drawn from an IMF study (2009), which evaluated the experiences of countries that faced or are facing deflation problem.

[4/] The index calculation based on an unweighted average may fail to capture the varying importance of each variable. However, when calculating a weighted average, quite a lot of detailed data are required to determine the appropriate weights for different variables. Further studies will be conducted for this purpose. Nonetheless, in using the current index, the BOT uses discretion in interpreting the calculated average index, attaching greater importance to some variables that contribute significantly to deflation risks, such as public inflation expectations.

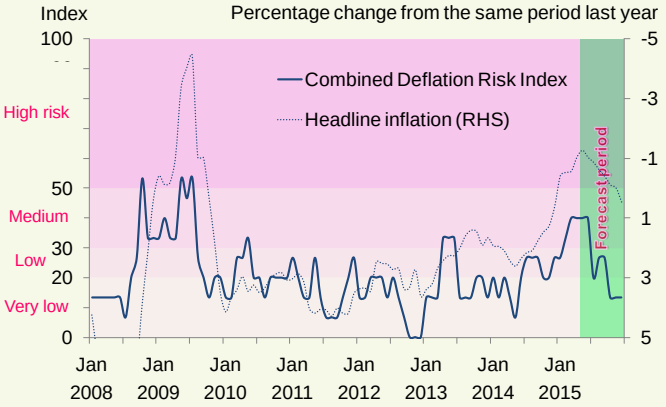
higher indicating high probability of deflation. To test whether the index can accurately captures risk of deflation in practice, the index for Japan over the 2009-2012 period, where the Japanese economy was in deflation, was calculated. The index had a value greater than 50, with almost all categories of factors signifying higher deflation risks **(Chart 1)**.

Chart 1 Japan's Combined Deflation Risk Index



Source: Calculations by Bank of Thailand

Chart 2 Thailand's Combined Deflation Risk Index



Source: Calculations by Bank of Thailand

In the case of Thailand, the index suggests that the probability of deflation in Thailand in 2015 is low. The index edged up in 2015 from lower energy and raw food prices, causing negative headline inflation. However, **our assessment of the horizon until the end of the year points to lower probability of deflation (Chart 2)** in line with higher costs. The effects of the high base of oil prices will gradually subside, while global oil prices is expected to rise somewhat in the second half of this year. In the meantime, domestic raw food prices are projected to pick up after the oversupply of meat and eggs begins to abate. Demand-side inflationary pressure is also projected to slowly increase as the economy recovers. Nonetheless, slower- or weaker-than-expected economic recovery could contribute to higher deflation risks ahead. Hence, the MPC continues to closely monitor Thailand's economic and financial developments, in particular those factors affecting private consumption including income, wages and employment, as well as consumer and business confidence.

Monetary and Financial Stability

3. Monetary and Financial Stability

Policy interest rate reductions at the March and April 2015 meetings led to more accommodative monetary conditions, with lower money market rates and commercial banks' lending rates. The baht weakened following the MPC rate decisions and the announcement of BOT's measures on capital flows relaxation under Capital Account Liberalization Master Plan, and the expectation of the imminent Fed's rate hike. Meanwhile, bank credits showed signs of modest improvement.

Financial positions of households and businesses, particularly small businesses, were adversely affected by slow economic recovery, leading to the deterioration in loan repayment abilities. Excess supply in real estate rose slightly due to stagnant purchasing power. Looking ahead economic recovery and accommodative monetary conditions, should contribute to healthier financial standings of households and businesses. Commercial banks maintained strict lending standards in the face of rising NPLs. Nevertheless, sufficient levels of loan loss provision and capital base should help maintain banking sector stability amid a decline in loan quality.

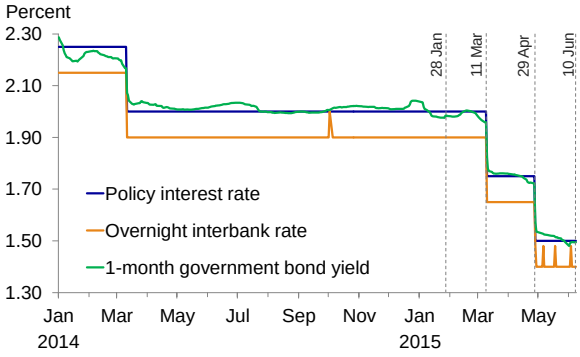
3.1 Financial markets

Monetary conditions became more accommodative following the two consecutive policy rate reductions at the March and April 2015 meetings. Money market rates fell immediately after the policy decisions, while government bond yields fell only initially before rising afterwards in tandem with global financial markets.

Money and bond markets

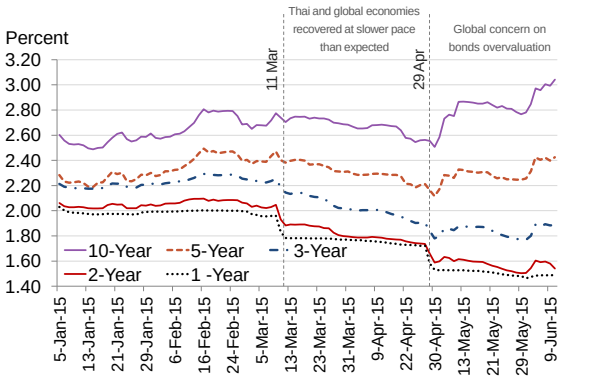
Monetary conditions eased after two reductions in policy interest rate on March 11 and April 29, 2015. Money market rates dropped immediately following both reductions **(Chart 3.1).** Similarly the yields on short-term government bonds

Chart 3.1 Money market interest rates



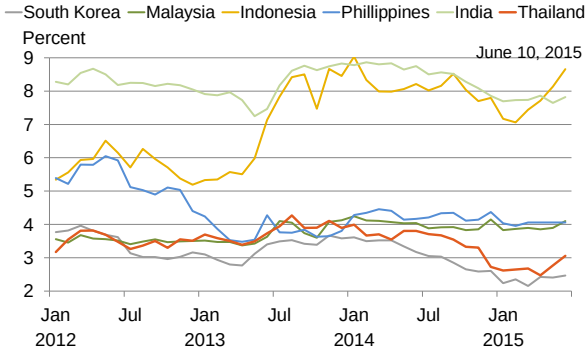
Source: Bank of Thailand and the Thai Bond Market Association (Thai BMA)

Chart 3.2 Government bond yields in 1 to 10-year sector



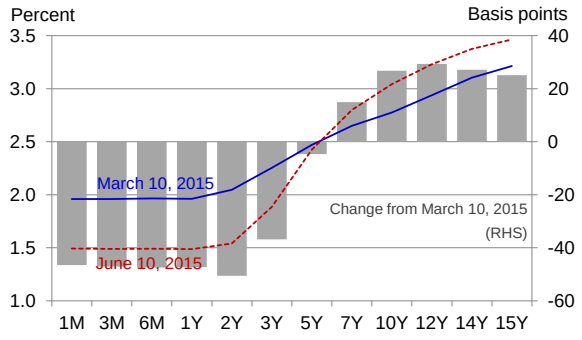
Source: Bank of Thailand and the Thai Bond Market Association (Thai BMA)

Chart 3.3 10-year government bond yields in regional countries



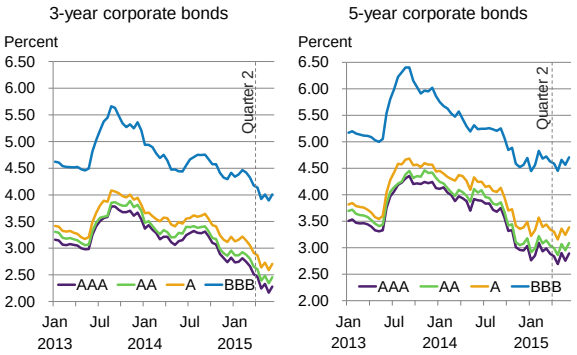
Source: Bloomberg

Chart 3.4 Changes in government bond yield curves



Source: Bank of Thailand and the Thai Bond Market Association (Thai BMA)

Chart 3.5 Thai corporate bond yields



Source: Thai Bond Market Association (Thai BMA)

(up to-2-year sector) declined due to the MPC's decisions, weaker economic momentum, as well as persistently subdued inflation **(Chart 3.2).**

In May 2015, however, yields on government bonds of medium to long maturities rose rapidly **(Chart 3.2),** in line with other regional bond yields **(Chart 3.3)**, following comments by the Fed's Chairwoman on possible overvaluation of the U.S. equities and treasury bonds. The remark immediately triggered sell-offs of assets across emerging markets. In sum, government bond yields up to the 5-year sector declined, whereas those of longer maturities increased from the pre-March 11, 2015 meeting levels **(Chart 3.4).** As a result, funding costs of short-term corporate bonds fell steadily, unlike that of medium to long-term ones, which edged up from the previous quarter **(Chart 3.5).**

Equity market

The equity market became less overheated compared to the previous quarter, due to government measures to curb speculation as well as sluggish economic growth. Although valuations of small-capitalization stocks remained elevated compared to historical average, risk to financial stability was deemed limited.

Compared to the first quarter, activities in the equity market became less overheated during the first two months of the second quarter, in line with sluggish economic activity. The Stock Exchange of Thailand (SET) index remained steady, while the Market for Alternative Investment (MAI) index fell. In April and May 2015, retail investors were net sellers in the SET market **(Chart 3.6)**, and they also reduced new purchases in the MAI **(Chart 3.7)**. On the other hand, foreign investors, who were net sellers in the first quarter, became net buyers in the SET during the second quarter.

Furthermore, measures implemented by the Securities and Exchange Commission (SEC) and Stock Exchange Thailand (SET), including Trading Alert List and Turnover List, helped curb the overheating of the market, as indicated by the continued decline in Price- Earnings (P/E) Ratio **(Chart 3.8).**

Despite these measures, the underpricing of risks among investors could pose a threat to financial stability. The Forward Price-Earnings (Forward P/E) Ratio[1/] remained substantially above historical averages especially in the MAI **(Table 3.1)**, although it had moderated compared to the end of 2014. Because stocks with high P/E ratio were mostly small-capitalization stocks, their total market capitalization was not large. Moreover, trading is funded mostly by cash, with low levels of margin loan. Therefore, these development posed limited risks to the overall financial stability.

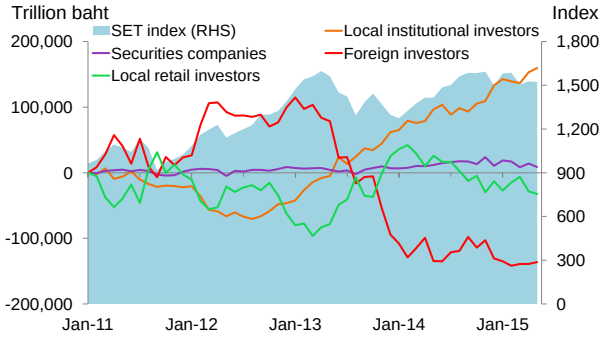
Table 3.1 Forward P/E ratio* in the SET and MAI (times)

Forward P/E Ratio (Times)	10-year average	2014				2015		
		Q1	Q2	Q3	Q4	Q1	Apr	1-25 May
SET	12.0	12.3	13.8	15.5	16.2	15.0	15.3	15.1
MAI	13.8	21.7	26.7	37.2	36.0	31.1	22.9	21.4

Note: Forward P/E Ratio is the current price divided by 1-year expected earnings per share
Source: Bloomberg and calculations by the Bank of Thailand

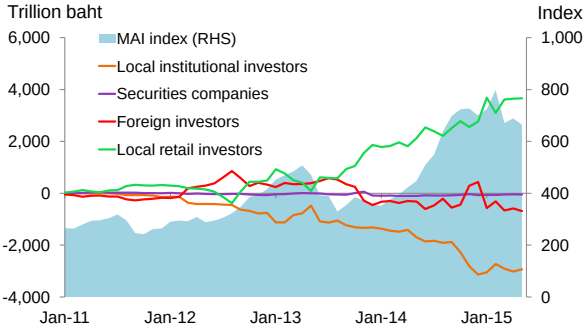
It is worth noting that domestic institutional investors and mutual funds were net buyers in the SET throughout the first half of this year. This can be explained in part by the welcome development of more efficient asset management by the general public. But this was likely also a result of growing search for yield behavior amid the prolonged

Chart 3.6 The Stock Exchange of Thailand index and net buy classified by investor types



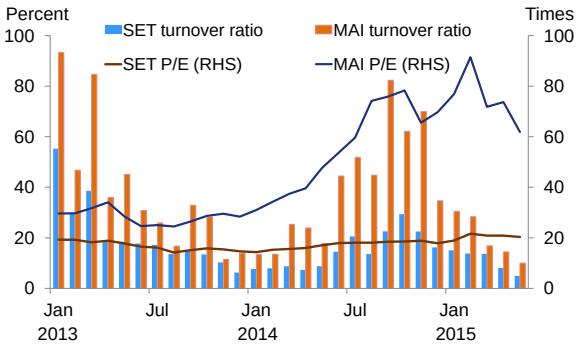
Source: The Stock Exchange of Thailand (latest data on May 25, 2015)

Chart 3.7 The Market for Alternative Investment (MAI) and net buy classified by investor types



Source: The Stock Exchange of Thailand (latest data on May 25, 2015)

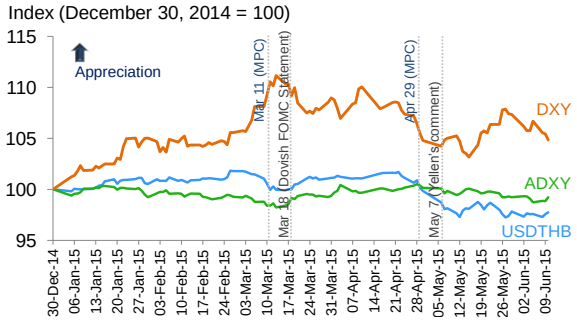
Chart 3.8 Turnover ratio in SET and MAI



Source: The Stock Exchange of Thailand (latest data on May 25, 2015)

[1/] P/E (Price-Earnings Ratio) is the ratio between current price and average net profit of the past 4 quarters. On the other hand, Forward P/E (Forward Price-Earnings Ratio) compares current price to 1-year expected earnings per share.

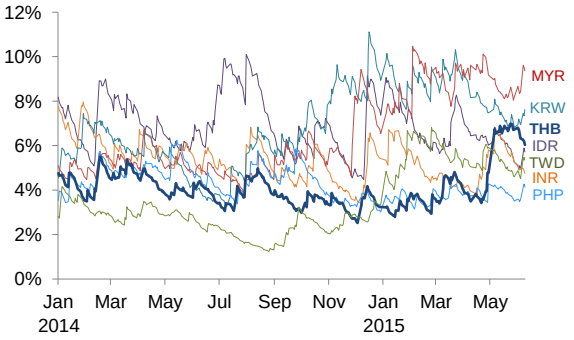
Chart 3.9 Movements in USDTHB and exchange rate indices



Note: DXY is an index of USD compared to a basket of EUR JPY GBP CAD SEK and CHF
ADXY is an index of Asian currencies (CNY KRW SGD HKD INR TWD THB MYR IDR and PHP) against USD

Source: USDTHB index from the Bank of Thailand, DXY and ADXY from Bloomberg

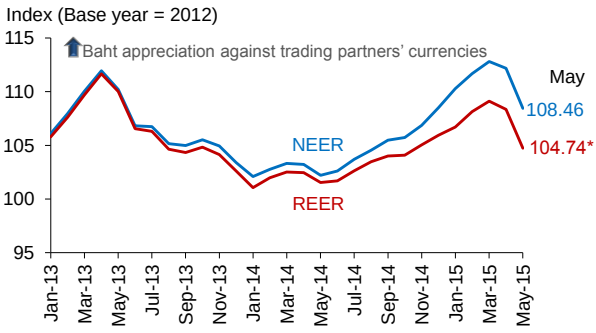
Chart 3.10 Volatility of the baht and regional currencies



Note: Volatility is computed by Exponentially Weighted Moving Average (EWMA) method

Source: The Bank of Thailand and Reuters

Chart 3.11 NEER and REER



Note: *Preliminary data using NEER from May 2015, and inflation from April 2015 as a proxy for May 2015 inflation

Source: Bank of Thailand

period of low-interest environment. Trading activities in equities and other asset classes, as well as their implications on financial stability, will continue to be closely monitored going forward.

Foreign exchange market

The Thai baht against U.S. dollar exchange rate stabilized in April 2015, before weakening steadily starting from the end of the month **(Chart 3.9)** as a result of the unexpected rate reduction on April 29, 2015. The ensuing announcement of BOT's measures on capital flows relaxation on April 30, 2015 also contributed to the continued weakening of the baht.

In May, asset sell-offs by non-residents across emerging economies, including Thailand, led to massive net outflows from the bond market. As a result, the baht depreciated further against U.S. dollar, closing at 33.66 baht per U.S. dollar on June 10, 2015, a 3.3 percent depreciation from the end of the first quarter. Also, the baht volatility surged compared to those of regional currencies during the same period **(Chart 3.10).**

Depreciation of the baht against major and most regional currencies led to a decline in Nominal Effective Exchange Rate (NEER) and Real Effective Exchange Rate (REER) in May 2015. In particular, the NEER fell by 2.8 percent from the last quarter **(Chart 3.11)**

3.2 Financial institutions

Following the monetary policy easing in March and April 2015, private sector's funding costs and returns on deposits declined. Commercial banks' benchmark loan rates and deposit rates dropped, although the degree to which loan rates responded to the policy rate cut on April 29, 2015 was more limited and slower than past averages. Meanwhile, credit expansion showed signs of some improvements. On the other hand, deposit growth continued to be subdued as ample liquidity amongst financial institutions reduced the competition for deposits.

Interest rates, credits, and deposits

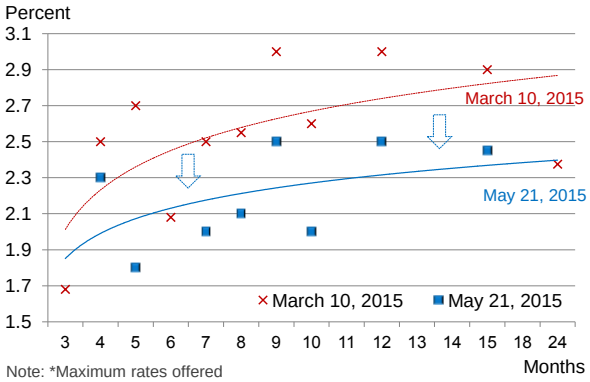
Funding costs of commercial bank loans declined following lower policy interest rate, especially after the March 11, 2015 decision. The majority of commercial banks reduced their Minimum Lending Rates (MLR) within the two weeks following the policy decision, although other benchmark rates, including Minimum Retail Rate (MRR), were kept constant **(Table 3.2).** Actual funding costs, however, would depend on each loan applicant's credit risk while commercial banks tighten their lending standards for both business and household.

The degree with which loan rates followed lower policy rate at the April 29, 2015 meeting was relatively slow and limited only to a few large commercial banks. This was because most banks were concerned over the outlook of bank earnings amid weaker-than-expected economic prospect and rising NPLs. More recently, however, the lending rates to creditworthy retail borrowers (MRR), offered by some large commercial banks, have been adjusted lower in line with the policy rate reductions.

Chart 3.2 Benchmark loans and deposits rates of commercial banks*								
Percent	2013	2014				2015		
		Q1	Q2	Q3	Q4	Q1	Apr 30	Jun 9
12-month deposits								
Average of the 4 largest commercial banks**	2.23	1.74	1.73	1.73	1.73	1.53	1.53	1.50
Average of remaining banks	2.61	2.35	2.22	2.19	2.22	1.93	1.90	1.77
Minimum lending rate (MLR)								
Average of the 4 largest commercial banks	6.84	6.75	6.75	6.75	6.75	6.63	6.63	6.51
Average of remaining banks	7.62	7.53	7.51	7.45	7.44	7.37	7.34	7.27
Minimum retail rate (MRR)								
Average of the 4 largest commercial banks	7.96	8.02	8.05	8.08	8.08	8.08	8.09	7.86
Average of remaining banks	8.77	8.70	8.70	8.63	8.63	8.62	8.61	8.54

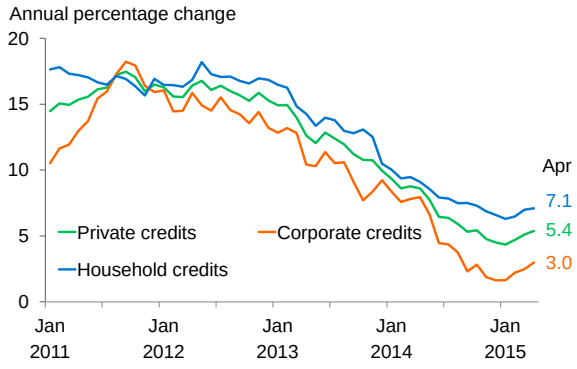
Note: *Benchmark rates averaged across commercial banks at the end of the period
** Four largest commercial banks namely BBL, KTB, KBANK and SCB
*** Since August 2014, Bank of China (Thailand) became a domestically-registered commercial bank. Therefore, its interest rates are included in the Average of remaining banks rates in the table above.

Chart 3.12 Special deposit rates*



Note: *Maximum rates offered
Source: Bank of Thailand

Chart 3.13 Other Depository Institutions' private credits



Source: Bank of Thailand

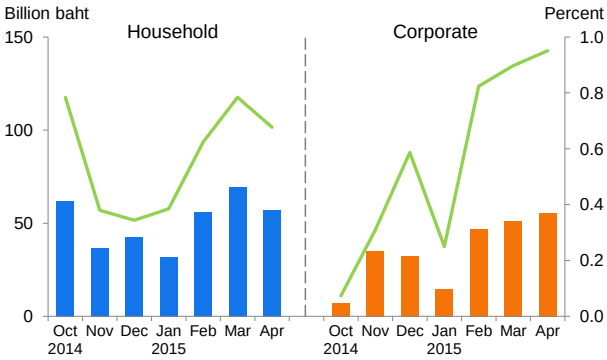
Meanwhile, commercial banks reduced their deposit rates in line with the fall in the policy rate **(Chart 3.12).** The reduction was also partly driven by ample liquidity among banks stemming from sluggish demand for credits. A more detailed look revealed that most banks cut the interest rates on fixed deposit and special deposit products, while leaving those on current deposit and savings accounts unchanged, both of which constitute the majority of total deposits.

Private credit growth accelerated slightly in 2015 Q1, up to 5.4 percent from 4.5 percent in the last quarter of 2014 **(Chart 3.13).** Signs of improvements in credit issuance became evident for both household and business, although growth in business loans was limited to just a few sectors **(Chart 3.14)**.

Credits to the private sector by depository institutions[2/] grew from 2014 Q4. However, new loan issuance was largely working capital loans, whereas fixed assets loans continued to slow in line with weak private investment. New issuances were also concentrated in a few business sectors, likely a result of transitory factors. For example, anticipated rise in the excise tax brought forward production and hence accelerated credit demands by the food and beverages industry in February. With regards to the capital markets, businesses obtained more funding via the bond market compared to the previous quarter **(Chart 3.15)**. The fall in short-term yields following policy rate cuts in March and April 2015 spurred the corporate bond issuance in the 1-3 year sector. Similarly, equity issuances rose sharply in March 2015, led by construction materials, real estate, and transportation sectors.

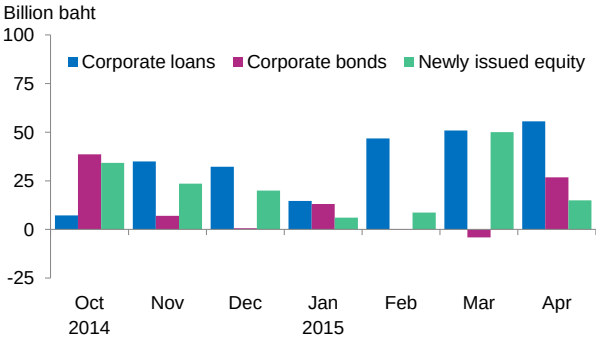
New issuance of household credits also expanded somewhat compared to the last quarter of 2014, driven partly by credits that Specialized Financial Institutions extended to village funds and farmers under the Ministry of Finance's initiatives. But given subdued economic recovery and uncertainty over future income, consumers remained cautious with regards to their spending. Thus, there is great uncertainty over the degree which consumption spending will benefit from these new credits. Meanwhile, household credit approval rate dropped from the previous quarter

Chart 3.14 New private credits
(Changes in credit balance, seasonally adjusted)



Source: Bank of Thailand

Chart 3.15 Corporate financing*

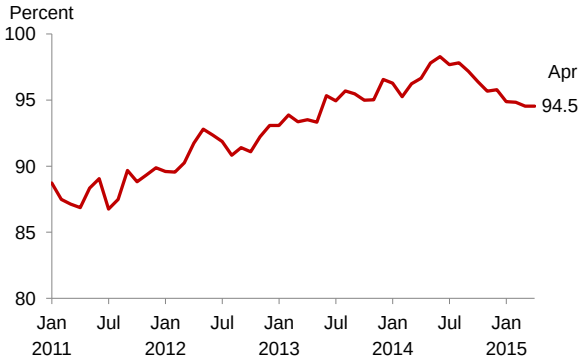


Note: *Newly issued equities, change in corporate loans (seasonally adjusted) and corporate bonds

Source: Bank of Thailand and Thai BMA

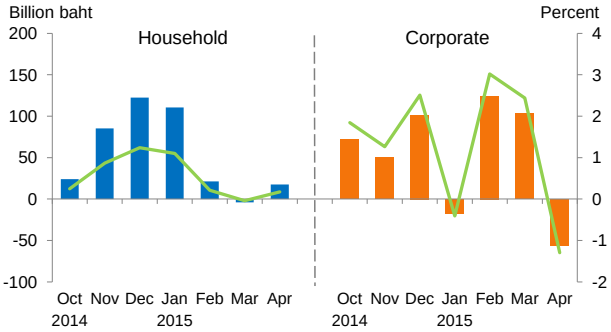
[2/] Depository institutions include domestically-registered commercial banks, branches of foreign banks, international banking facilities (ceased operations in 2006), financial companies, specialized financial institutions, savings cooperatives, and money market mutual funds, excluding the Bank of Thailand.

Chart 3.16 Loans to deposits and B/E ratio of commercial banks



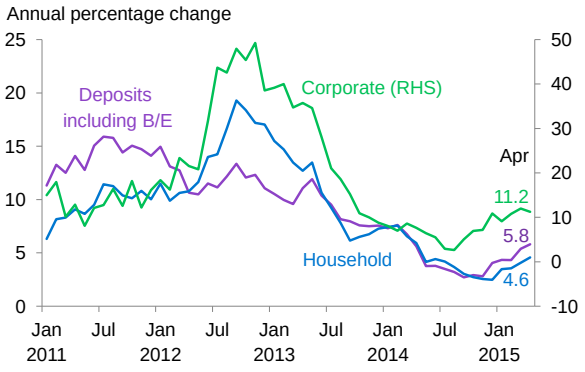
Source: Bank of Thailand

Chart 3.17 New deposits*



Note: *Change in outstanding deposits at all depository institutions (excluding the Bank of Thailand). This includes neither transfers of deposits within and among banks, nor rollovers of deposits.

Chart 3.18 Other Depository Institutions' deposits



Source: Bank of Thailand

as some financial institutions tightened their credit standards.

Looking ahead, the latest Senior Loan Officers' Survey indicates that demand for business credits will pick up slightly. Most of the increased demand is for working capital in sectors that mainly produce for the domestic market, as well as tourism-related industries, such as food and beverages, hotels and restaurants, and wholesale and retail trade. Household credits are also expected to grow, especially those for other consumption categories. Nevertheless, given the concerns over weak economic prospects and worsening credit quality, financial institutions will likely remain cautious in lending.

Since 2014 Q4, the liquidity position of commercial banks has increased, as indicated by loan to deposit (including bills of exchange (B/E)) ratio falling from 98.3 percent in June 2014 to 94.5 percent at the end of April 2015 **(Chart 3.16)**[3/]. That, coupled with slower-than-expected economic recovery, led to a decline in competition for deposits among banks, resulting in lower interest rates on special deposit products. As a result, the amount of new deposits shrank in 2015 Q1 and in April **(Chart 3.17)**. Yet deposits and bills of exchange (B/E) for other depository institutions rose by 5.8 percent from the same period last year, and by 4.0 percent from the previous quarter **(Chart 3.18).**

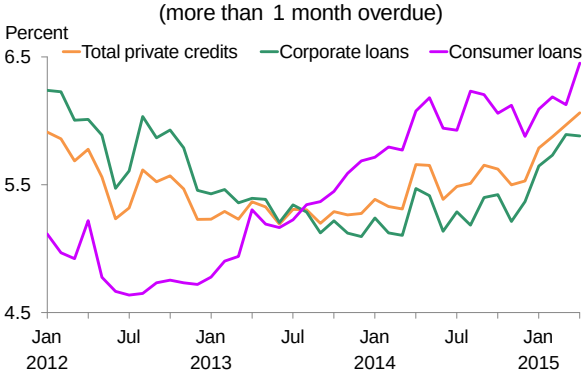
[3/] Although credits accelerated and competition for deposits among financial institutions eased, loan to deposit including bills of exchange (B/E) ratio declined from the previous quarter because deposits constitute a large base.

Stability of financial institutions

Earnings and stability of financial institution system remained resilient, although loan quality deteriorated somewhat from the end of 2014 due to the slow economic recovery. Nevertheless, commercial banks have maintained high level of loan loss provision and capital base that would serve as cushion against economic uncertainties.

Overall loan quality worsened, reflected by the delinquency and NPL Ratio (those loans with more than one month overdue) that climbing to 6.0 percent **(Chart 3.19).** Worsening loan quality were detected among the SME credits and all types of consumer credits **(Table 3.3).** Nonetheless, sound risk management and high level of provision have continued to ensure the stability of financial institutions. The ratio of Actual to Regulatory Loan Loss Provision Ratio stood at 165.9 percent, while Capital Adequacy Ratio was at 16.6 percent, both indicating strong financial position which should be able to absorb shocks from deteriorating loan quality.

Chart 3.19 Delinquency and NPL Ratio
(more than 1 month overdue)



Note: Excluding loans to government and Interbank loans
Source: Bank of Thailand

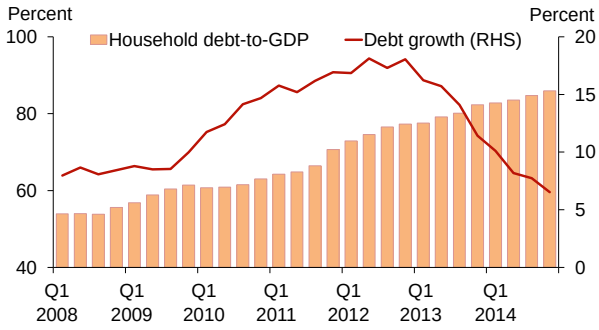
3.3 Non-financial sectors

Household sector

Sluggish economic recovery resulted in lower household income, thus weakening household financial positions. Therefore, consumers have become more guarded with respect to spending and incurring new debts. More cautious lending standard by financial institutions also contributed to the slowdown in household debt accumulation. Furthermore, repayment ability among households with high debt outstanding was negatively impacted by the weak economy, reflected by the rising

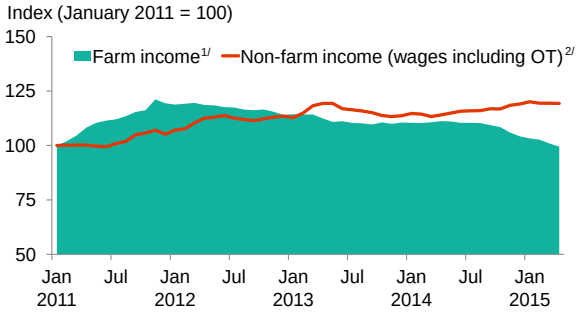
delinquency and NPL ratios across all types of consumer loans.

Chart 3.20 Household debt[1/] to GDP[2/]



Note: [1/] Loans to households by financial institutions
[2/] Computed using fix-based GDP (base year = 1988)
Source: Bank of Thailand

Chart 3.21 Household income



Note: [1/] Seasonally adjusted, 12-month moving average
[2/] Seasonally adjusted, 3-month moving average
Source: Office of Agricultural Economics and National Statistical Office, calculations by the Bank of Thailand

Weak economic recovery, coupled with prospects of falling income, caused households to be cautious in spending and incurring new debts. Household credits, therefore, dropped from 7.7 percent at the end of 2014 Q3 to 6.5 percent at the end of 2014. With household debt grew faster than GDP growth, household debt-to-GDP ratio, therefore, continued to edge up to 85.9 percent at the end of 2014 **(Chart 3.20).**

Slow economic recovery also lowered household income and undermined consumer confidence. Income in the agricultural sector continued to contract following the downward trend in prices of agricultural products. Similarly, non-agricultural income (wages including overtime pay) dropped slightly **(Chart 3.21)**. Taken together, repayment ability of households declined, indicated by rising NPLs and delinquency ratios (share of loans overdue by over 1 month) which grew from 5.9 percent at the end of 2014 to 6.1 percent at the end of 2015 Q1 **(Table 3.3)**. Overall, the decline in quality of personal loans and credit card loans, has been observed across all income groups.

In the coming months, the MPC will continue to closely monitor the impacts of economic activity on household income and repayment ability, particularly among the low-income and agricultural households whose debt burdens are more elevated compared to others.

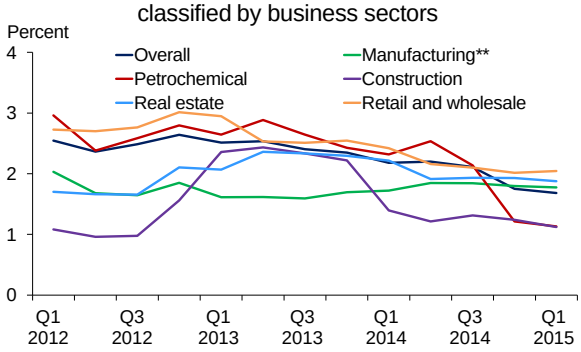
Corporate sector

Earnings of corporate sector declined slightly from the previous quarter. Moreover, financial standing of households and businesses, especially small businesses, were affected by slow economic recovery, causing a decline in repayment ability.

In 2015 Q1, earnings of listed companies showed signs of slowing down from the previous quarter, indicated by a decline in the Return on Asset ratio (ROA) **(Chart 3.22)**. In the real estate sector, sluggish economic recovery led to lower sales, while firms incurred surges in marketing and operational expenses. As a result, Net Profit Margin (NPM) for the real estate sector fell. **(Chart 3.23)**. Likewise, construction firms have not benefited significantly from government's infrastructure investment. Meanwhile, the petroleum sector continued to be undermined by the subdued global crude oil prices, resulting in the decline of the Asset Turnover Ratio (ATO) **(Chart 3.24)**. Nevertheless, their earnings were partly improved from the fact that stock losses and asset impairments had been realised in the previous accounting periods. On the other hand, growing tourism sector benefited hotels and restaurants, as well as transportation industries, particularly the airline industry. Declining oil prices also boosted the earnings of transportation group.

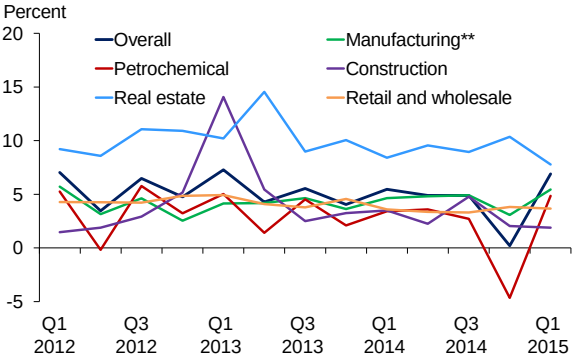
Despite these developments, risks to the stability of the corporate sector remain low, evidenced by the stable Debt-to-Equity ratio **(Chart 3.25)**. Overall the repayment ability fell slightly, indicated by a small drop in Interest Coverage Ratio (ICR), but at levels still considered robust **(Chart 3.26)**. A closer look at firms across different sizes (5 groups classified by sizes),

Chart 3.22 Return on Asset* (ROA) classified by business sectors



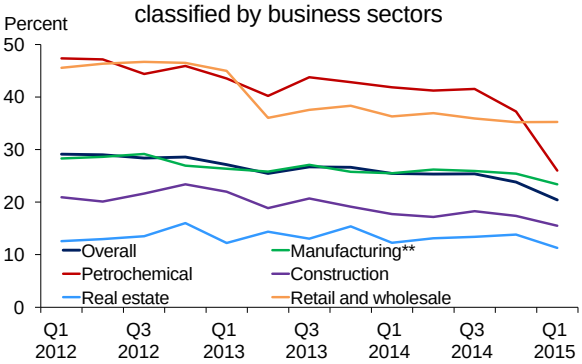
Note: *Average financial ratio of each sector
**Excluding petrochemical
Source: The Stock Exchange of Thailand. Calculations by the Bank of Thailand.

Chart 3.23 Net Profit Margin* classified by sectors



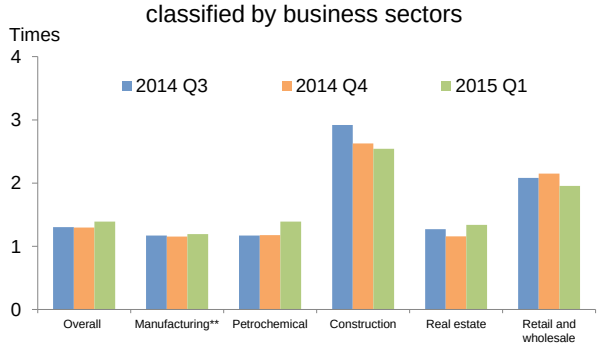
Note: *Average financial ratio of each sector
**Excluding petrochemical
Source: The Stock Exchange of Thailand. Calculations by the Bank of Thailand.

Chart 3.24 Asset Turnover Ratio* classified by business sectors



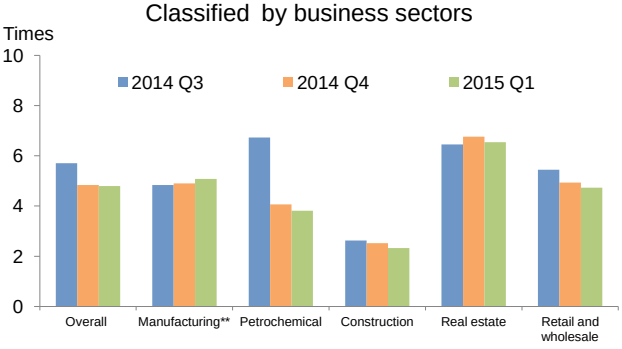
Note: *Average financial ratio of each sector
**Excluding petrochemical
Source: The Stock Exchange of Thailand,. Calculations by the Bank of Thailand.

Chart 3.25 Debt-to-Equity Ratio* classified by business sectors



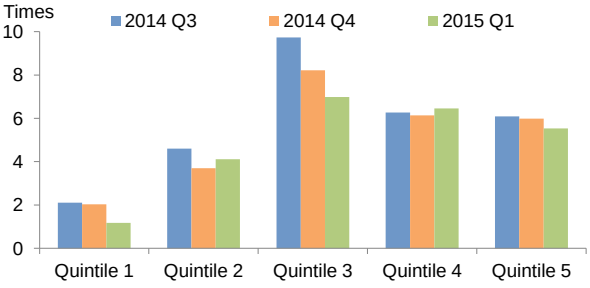
Note: *Average annualized financial ratio of each sector
**Excluding petrochemical
Source: The Stock Exchange of Thailand,. Calculations by the Bank of Thailand.

Chart 3.26 Interest Coverage Ratio* Classified by business sectors



Note: *Average annualized financial ratio of each sector
**Excluding petrochemical
Source: The Stock Exchange of Thailand. Calculations by the Bank of Thailand.

Chart 3.27 Interest Coverage Ratio* Classified by firm sizes**



Note: *Average annualized financial ratio of each sector
**Firm sizes are classified into quantiles using average annual sales.
Diagram shows the median of each quantile.
Source: The Stock Exchange of Thailand. Calculations by the Bank of Thailand.

however, revealed that medium to large listed companies maintained high repayment ability, with high ICR **(Chart 3.27)**. On the other hand, the smallest firm group exhibited sharp declines in ICR, signaling potential liquidity risks. If these small listed firms are taken to reflect the situations of the non-listed firms which tend to be small and medium sized firms, this may indicate the worsening liquidity positions and lower repayment ability of SMEs as a result of the lethargic economy as well. Thus, the MPC will continue to closely monitor this issue going forward **(Table 3.3).**

Real estate sector

Excess supply in the real estate sector continued to rise, particularly for condominiums, as consumer purchasing power was adversely affected by slow economic growth and banks' cautious stance towards new issuance of mortgage loans. Nevertheless, the real estate sector, which is dominated by large firms, possessed solid financial position and continued to adapt to the changing business conditions. Thus, the sector is likely to pose limited stability risk to overall financial system at the moment.

Demand in real estate market slowed markedly over the first five months of 2015, as a result of weak economic recovery, subdued household incomes, persistently elevated level of household debt, and more stringent lending standards applied to the sector by financial institutions. As such, most real estate firms postponed new project launches, although on balance the number of project launches rose because some large firms decided to relaunch shelved projects in April **(Chart 3.28).** The majority of these new projects were condominiums along mass transit rail routes, for which developers

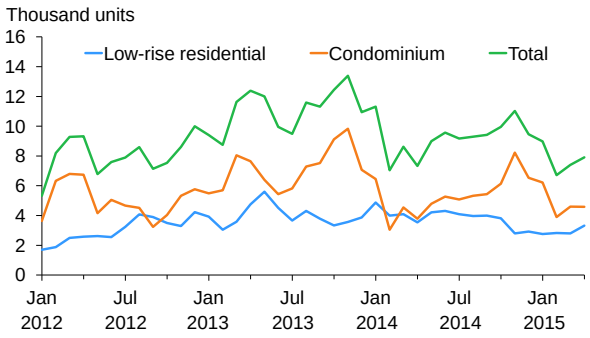
expected demand to bounce back in the near future. Furthermore, a number of firms repositioned themselves towards the middle to high-income consumers, whose purchasing power remaining relatively unaffected **(Chart 3.29).** Nevertheless, supply of new units still exceeded demand. The reservation rate of new residential units declined further to 20.9 percent in April 2015 from the end of 2014, well below the 2004-2014 average of 37.5 percent. Excess supply of real estate units, therefore, was expected to rise in the coming months.

However, since most real estate firms maintained solid financial standing, the risks to overall economic and financial stability of the slowdown in the sector should be limited. Retained Earnings to Total Assets Ratio **(Chart 3.30)** and Interest Coverage Ratio **(Chart 3.31)** of the sector still indicated robust health. Moreover, the quality of pre-finance **(Chart 3.32)** and post-finance loans declined only marginally, while banks remained cautious with respect to lending. Ample levels of Allowance for Doubtful Account and capital base should sufficiently deal with the risk of deteriorating loan quality.

Fiscal sector

Stability of the fiscal sector continued to be robust overall. Treasury cash balance was sufficient to buffer the impact of lower-than-projected tax revenue collection. Moreover, the current level of public debt remained low and does not pose threat to stability. Nevertheless, the level of public debt is likely to rise over the coming months as a result of transfers of debt incurred by Specialized Financial Institutions' quasi-fiscal activities, as well as other outstanding debt that the government owes to various funds and public enterprises. The fiscalization of such debt will raise public debt burden and thus

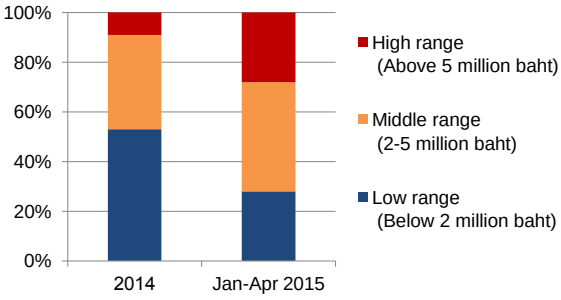
Chart 3.28 New residential project launches in Bangkok and its vicinities



Note: * 3-month moving average

Source Agency for Real Estate Affairs (AREA) and calculations by Bank of Thailand

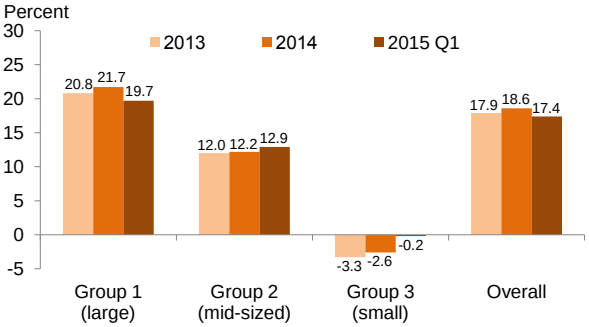
Chart 3.29 Shares of new residential project launches classified by price per unit*



Note: *Bangkok and vicinities only

Source: Real Estate Information Center (REIC)

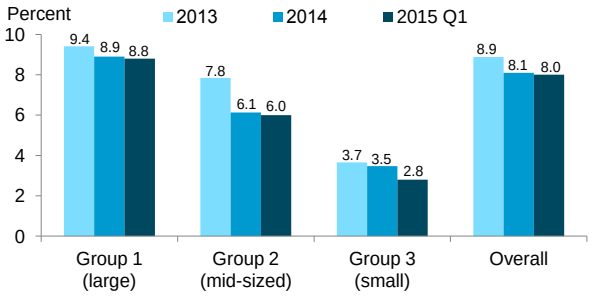
Chart 3.30 Retained earnings to total asset ratio*



Note: * Only listed companies are considered.
Group 1 (large), Group 2 (mid-sized), and Group 3 (small) constitute 75, 20, and 5 percent respectively of total real estate asset at the end of 2014 Q4.

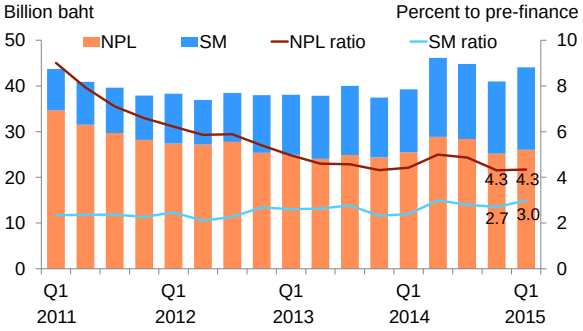
Source: Stock Exchange of Thailand and calculations by Bank of Thailand.

Chart 3.31 Interest Coverage Ratio*



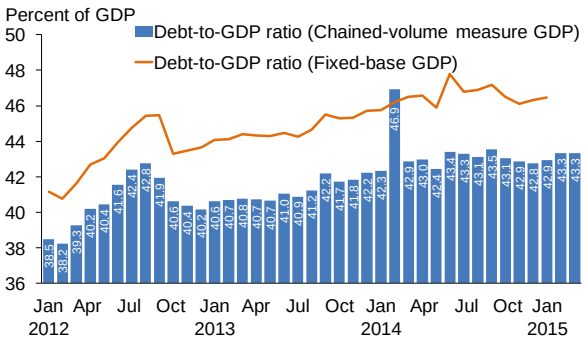
Note: *Considers only listed companies
Group 1 (large), Group 2 (mid-sized), and Group 3 (small) constitute 75, 20, and 5 percent respectively of total real estate asset at the end of 2014 Q4.
Source: Stock Exchange of Thailand and calculations by Bank of Thailand.

Chart 3.32 Mortgage loan quality of commercial banks (Pre-finance)



Source: Bank of Thailand

Chart 3.33 Public Debt to GDP ratio*



Note: (1) Chart shows calendar years, (2) Official figures for Debt-to-GDP ratio based on Chained-Volume Measure GDP are only available for February and March 2015. The figures prior to this period are calculated by the Bank of Thailand.
Source: Public Debt Management Office

will be one of the key issues that needs to be monitored going forward.

The outlook for fiscal sector remained stable, as reflected by a number of indicators. (1) Treasury budget balance had increased and was sufficient to meet necessary demand of liquidity as well as offset the projected shortfall of revenue collection this year, marking two consecutive years of revenue shortfall in line with economic slowdown. Treasury budget balance at the end of April 2015 stood at 126 billion baht, up from 116 billion baht in January 2015. The increase was a result of the fact that the government had borrowed more than needed to cover for monthly budget deficits. (2) **The ratio of public debt to GDP*/ grew slightly, but remained well below the fiscal sustainability threshold of 60 percent**. In March 2015, the ratio recorded at 43.3 percent, up from 42.8 percent in December 2014 **(Chart 3.33).** New borrowings that covered for budget deficits and those provisioned for the restructuring of debt due on May 22, 2015 contributed to the increase in the public debt to GDP ratio.

Going forward, a major risk factor that requires close monitoring is the potential surge in public debt following the debt transfer plan. First, debts incurred by SFIs for quasi-fiscal measures will be transferred to the government. Moreover, further outstanding debt burden includes the amounts that the government owes to various funds and public enterprises, such as rice subsidy

*/ Debt-to-GDP ratio is disseminated by the Public Debt Management Office of the Ministry of Finance. The current series is constructed using the new chained volume measure GDP, publicized on May 18, 2015. The ratios in this series are significantly smaller than those constructed using the former GDP series (with fixed base year).

scheme, crop price guarantee scheme, social security fund, and other state-owned enterprises. Nevertheless, new policy which requires existing revolving funds to return excess liquidity to the treasury, which amounts to 30 billion baht in total, would partly lessen the risk to fiscal stability emerging from the imbalance between revenue and spending.

Table 3.3 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2014				2015		
		Q1	Q2	Q3	Q4	Q1	Apr	May
1. Financial markets sector								
Bond market								
Bond spread (10 years-2 years)	1.3	1.5	1.4	1.2	0.9	0.7	0.9	1.2
Equity market								
SET Index (End of period)	1,497.7	1,376.3	1,485.8	1,585.7	1,497.7	1,505.9	1,526.7	1,496.1
Actual volatility (SET Index)[1/]	11.9	16.1	10.4	8.3	12.8	12.6	11.6	13.3
Price to Earnings Ratio (times)	17.8	15.6	17.9	18.4	17.8	20.9	20.9	20.3
FX market								
Actual volatility (baht) (%annualized)[2/]	4.0	4.5	4.1	3.8	3.4	3.7	3.9	6.7
Nominal Effective Exchange Rate (NEER)	104.3	102.7	102.7	104.6	107.0	111.6	112.2	108.5
Real Effective Exchange Rate (REER)	103.0	101.9	101.9	103.4	105.0	108.0	108.4	n.a.
2. Financial institutions sector[3/]								
Minimum lending rate (MLR)[4/]	6.8	6.8	6.8	6.8	6.8	6.6	6.6	6.5
12-month fixed deposit rate[4/]	1.7	1.7	1.7	1.7	1.7	1.5	1.5	1.5
Capital adequacy								
Regulatory capital to risk-weighted asset (%)		15.5	15.9	17.1	16.8	16.6		
Earnings and profitability								
Net profit (billion baht)		50.5	59.8	53.8	48.7	52.5		
Return on assets (ROA)		1.26	1.48	1.33	1.18	1.26		
Liquidity								
Loan to deposit and B/E		95.9	97.9	97.2	95.7	94.5		
3. Household sector								
Household debt to GDP (times)	85.9	82.8	83.5	84.7	85.9	n.a.		
Financial assets to debt (times)	2.1	2.0	2.0	2.0	2.1	n.a.		
NPL and delinquency ratio (%)								
Thai commercial banks :								
Consumer loans	5.9	5.8	5.9	6.2	5.9	6.1		
Mortgage loans	3.8	3.8	3.8	4.1	3.8	4.1		
Auto leasing	10.8	9.9	10.5	10.7	10.8	10.9		
Credit cards	5.4	5.3	5.6	6.1	5.4	6.5		
Other personal loans	5.0	4.9	5.0	5.6	5.0	5.3		
4. Non-financial corporate sector [5/]								
Operating profit margin (%)	3.9	5.5	4.9	4.9	0.2	6.9		
Debt to equity ratio (times)	1.3	1.3	1.3	1.3	1.3	1.4		
Income coverage ratio (times)	4.9	6.5	5.6	5.7	1.7	6.2		
Current ratio (times)	1.4	1.4	1.4	1.4	1.4	1.4		

Table 3.3 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2014				2015		
		Q1	Q2	Q3	Q4	Q1	Apr	May
5. Real estate sector								
The number of approved mortgages from banks (Bangkok and its vicinity)	62,839	12,880	16,315	17,345	16,299	11,564	4,026	
Single-detached and semi-detached houses	15,694	3,331	4,206	4,230	3,927	3,001	913	
Townhouses and commercial buildings	21,764	4,784	5,921	5,844	5,215	4,212	1,302	
Condominiums	25,381	4,765	6,188	7,271	7,157	4,351	1,811	
The number of new openings (Bangkok and its vicinity)	111,211	25,862	28,714	28,268	28,367	22,191	8,308	
Single-detached and semi-detached houses	18,933	4,754	5,657	5,349	3,173	3,231	1,232	
Townhouses and commercial buildings	26,980	7,499	7,261	6,611	5,609	5,172	2,044	
Condominiums	65,298	13,609	15,796	16,308	19,585	13,788	5,032	
Housing price index [6/]								
Single-detached houses (including land)	117.0	114.7	115.5	119.3	118.4	119.4	119.5	
Townhouses (including land)	130.2	125.7	128.6	134.2	132.5	135.8	137.0	
Condominiums	154.3	146.2	153.1	156.0	161.9	162.9	162.6	
Land	150.7	145.9	148.3	153.0	155.5	163.1	163.1	
6. Fiscal sector								
Public debt to GDP (%) [7/]	42.8	42.9	43.4	43.5	42.8	43.3	43.5	

[1/] Daily volatility (using exponentially weighted moving average method)

[2/] Annualized volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 4 largest Thai commercial banks

[5/] Only listed companies on SET (median)

[6/] Based on data of new approvals by commercial banks using hedonic regression method (January 2009 = 100)
(Due to the fact that the structure of the housing market has changed significantly, the Bank of Thailand is currently improving the price index to better reflect the structure change)

[7/] Revised data with new GDP calculated using chain volume measure method